Model Architecture: gemma-3-12b-it

Generated by sg_modelload

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1 Model Overview

Model name	gemma-3-12b-it
Transformers class	`Gemma3ForConditionalGeneration`
Path	`/models/gemma-3-12b-it`
Total parameters	12,187,325,040 (12.19B)
Weight dtype	bfloat16
Hidden size d	3,840
Decoder layers L	48
Query heads H	16
Key/value heads H_{kv}	8
Head dim d_h	256
MLP width d_{ff}	15,360
Vocabulary V	262,208
Max context length	131,072
Sliding window w	1,024
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	$\mathrm{GELU}_{\mathrm{tanh}}$
Vision encoder: layers	27
hidden size / heads	1152 / 16

How to read this document. Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Notation

B, T	batch size, sequence length (tokens)
d	hidden size (3840)
V	vocabulary size (262,208)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
$\mathrm{RMSNorm}(x)$	$\gamma \odot x / \sqrt{\mathrm{mean}(x^2) + \epsilon}$ (learned scale γ)
$\mathrm{LN}(x)$	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 The Model as a Function

Input: token ids $x \in \{0, \ldots, V-1\}^{B \times T}$. Output: next-token logits $y \in \mathbb{R}^{B \times T \times V}$. The 48 decoder layers are written with composition powers, so a block that repeats n times appears once as $\mathcal{B}^{\circ n}$.

$$
\begin{aligned}
h_0 &= \mathrm{Merge}\Big(\mathrm{Embed}(x) \cdot \sqrt{d},\ \mathrm{Proj}_{\mathrm{img}}\big(\mathrm{Enc}_{\mathrm{V}}(\mathrm{img})\big)\Big) \\
h_L &= \Big(\big(\mathcal{B}_{\mathrm{F}} \circ \mathcal{B}_{\mathrm{S}}^{\circ 5}\big)^{\circ 8}\Big)(h_0) \\
y &= W_E^{\top}\ \mathrm{RMSNorm}_{\mathrm{final}}(h_L)
\end{aligned}
$$

Merge puts the projected image/audio vectors at the placeholder-token positions of the text embedding sequence. The encoders are themselves repeated stacks:

$$
\mathrm{Enc}_{\mathrm{V}} = \mathrm{Post} \circ \mathcal{B}_{\mathrm{V}}^{\circ 27} \circ \mathrm{PatchEmbed}
$$

2.1 Layer pattern

Symbol	Block type	Count	Params / block	Layer indices
$\mathcal{B}_{\mathrm{S}}$	Sliding-window attention block	40	224,148,992	0–4, 6–10, 12–16, 18–22, 24–28, 30–34, 36–40, 42–46
$\mathcal{B}_{\mathrm{F}}$	Full (global) attention block	8	224,148,992	5, 11, …, 47 (every 6th)
$\mathcal{B}_{\mathrm{V}}$	Vision encoder block	27	15,239,504	0–26

Language-model decoder (48 layers). layers 0–47: $[\mathcal{B}_{\mathrm{S}}\times 5, \mathcal{B}_{\mathrm{F}}]$ repeated 8×.

Vision encoder (27 layers). layers 0–26: $[\mathcal{B}_{\mathrm{V}}]$ repeated 27×.

2.2 Block functions

$\mathcal{B}_{\mathrm{S}}$ — **Sliding-window attention block.**

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention}\\
\mathcal{B}(h) &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP}\\
Q &= W_Q\,u && H=16 \text{ heads of size } d_h=256\\
K,\ V &= W_K\,u,\quad W_V\,u && H_{kv}=8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\,Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal + sliding window of } w=1024\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H]\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width } 15360
\end{aligned}
$$

$\mathcal{B}_{\mathrm{F}}$ — **Full (global) attention block.**

$$
\begin{aligned}
a &= h + \mathrm{RMSNorm}_{\mathrm{pa}}\big(\mathrm{Attn}(\mathrm{RMSNorm}_{\mathrm{in}}(h))\big) && \text{norm before AND after attention}\\
\mathcal{B}(h) &= a + \mathrm{RMSNorm}_{\mathrm{pf}}\big(\mathrm{FFN}(\mathrm{RMSNorm}_{\mathrm{pre}}(a))\big) && \text{norm before AND after the MLP}\\
Q &= W_Q\,u && H=16 \text{ heads of size } d_h=256\\
K,\ V &= W_K\,u,\quad W_V\,u && H_{kv}=8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\,Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H]\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\mathrm{GELU}_{\tanh}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width } 15360
\end{aligned}
$$

$\mathcal{B}_{\mathrm{V}}$ — **Vision encoder block.**

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{LN}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{LN}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\,u + b_Q && H=16 \text{ heads of size } d_h=72\\
K,\ V &= W_K\,u + b_K,\quad W_V\,u + b_V && H_{kv}=16 \text{ key/value heads}\\
S_h &= s\,Q_h K_h^{\top} + M && \text{mask } M\text{: bidirectional}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_h\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H] + b_O\\
\mathrm{FFN}(u) &= W_2\ \mathrm{GELU}_{\tanh}(W_1 u + b_1) + b_2 && \text{MLP, width } 4304
\end{aligned}
$$

3 Forward Pass in Pseudocode

Algorithm 1 gemma-3-12b-it: full forward pass

Require: token ids x of shape $B \times T$; optional images / audio

$h \leftarrow \mathrm{Embed}(x) \cdot \sqrt{d}$ ▷ look up one vector per token: $B{\times}T{\times}3840$
if image input present **then**
  $e \leftarrow \mathrm{Enc}_{\mathrm{V}}(\text{image})$ ▷ vision encoder: 27 layers, Algorithm 2
  $h[\text{image token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 3840
end if
for $r \leftarrow 0$ **to** 7 **do** ▷ layers 0–47: pattern of 6 repeated 8 times
  for $j \leftarrow 1$ **to** 5 **do**
    $h \leftarrow \mathcal{B}_{\mathrm{S}}(h)$ ▷ Sliding-window attention block
  end for
  $h \leftarrow \mathcal{B}_{\mathrm{F}}(h)$ ▷ Full (global) attention block
end for
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$
$y \leftarrow h\,W_E^{\top}$ ▷ logits $B{\times}T{\times}262,208$, embedding matrix re-used
return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Vision encoder

$h \leftarrow \mathrm{PatchEmbed}(\text{pixels}) + \text{position embedding}$ ▷ one vector per image patch
for $\ell \leftarrow 0$ **to** 26 **do**
  $h \leftarrow \mathcal{B}_{\mathrm{V}}(h)$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 2
return h

Algorithm 3 Block $\mathcal{B}_{\mathrm{S}}(h)$ — Sliding-window attention block

Require: h of shape $B \times T \times 3840$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^{\top}$ ▷ $B{\times}T{\times}16{\times}256$ (16 query heads)
$K \leftarrow uW_K^{\top};\ V \leftarrow uW_V^{\top}$ ▷ $B{\times}T{\times}8{\times}256$ (8 key/value heads, each shared by 2 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^{\top} + M)$ ▷ mask: causal + only the last 1024 tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}256$
$m \leftarrow O\,W_O^{\top}$ ▷ concatenate heads, project back to $B{\times}T{\times}3840$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^{\top}) \odot uW_{\mathrm{up}}^{\top}\big)W_{\mathrm{down}}^{\top}$ ▷ expand to 15360, gate, shrink back to 3840
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
return h

Algorithm 4 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 3840$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}16{\times}256$ (16 query heads)
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 key/value heads, each shared by 2 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}256$
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}3840$
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-attention norm, then add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{GELU}_{\tanh}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 15360, gate, shrink back to 3840
$h \leftarrow r + \mathrm{RMSNorm}(m)$ ▷ post-feed-forward norm, then add residual
return h

Algorithm 5 Block $\mathcal{B}_{\mathrm{V}}(h)$ — Vision encoder block

Require: h of shape $B \times T \times 1152$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{LN}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top + b_Q$ ▷ $B{\times}T{\times}16{\times}72$ (16 query heads)
$K \leftarrow uW_K^\top + b_K;\ V \leftarrow uW_V^\top + b_V$ ▷ $B{\times}T{\times}16{\times}72$ (16 key/value heads)
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: every patch/frame sees every other
$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}72$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}1152$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{LN}(h)$
$m \leftarrow \mathrm{GELU}_{\tanh}(uW_1^\top + b_1)W_2^\top + b_2$ ▷ expand to 4304, shrink back to 1152
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

Weight	Type	Shape	Params
`model.vision_tower.embeddings.patch_embedding.weight`	Conv2d	$1152 \times 3 \times 14 \times 14$	677,376
`model.vision_tower.embeddings.patch_embedding.bias`	Conv2d	1152	1,152
`model.vision_tower.embeddings.position_embedding.weight`	Embedding	4096×1152	4,718,592
`model.vision_tower.post_layernorm.weight`	LayerNorm	1152	1,152
`model.vision_tower.post_layernorm.bias`	LayerNorm	1152	1,152
`model.multi_modal_projector.mm_input_projection_weight`	Gemma3MultiModal Projector	1152×3840	4,423,680
`model.multi_modal_projector.mm_soft_emb_norm.weight`	Gemma3RMSNorm	1152	1,152

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Weight	Type	Shape	Params
`model.language_model.embed_tokens.weight`	Gemma3TextScaledWordEmbedding	262208×3840	1,006,878,720
`model.language_model.norm.weight`	Gemma3RMSNorm	3840	3,840
`lm_head.weight` (tied to `model.language_model.embed_tokens.weight`)	Linear	262208×3840	0 (shared)

4.2 Language-model decoder: `model.language_model.layers`

Table 3: $\mathcal{B}_{\mathrm{S}}$: Sliding-window attention block — layers 0–4, 6–10, 12–16, 18–22, 24–28, 30–34, 36–40, 42–46; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×3840	15,728,640
`self_attn.k_proj.weight`	Linear	2048×3840	7,864,320
`self_attn.v_proj.weight`	Linear	2048×3840	7,864,320
`self_attn.o_proj.weight`	Linear	3840×4096	15,728,640
`self_attn.q_norm.weight`	Gemma3RMSNorm	256	256
`self_attn.k_norm.weight`	Gemma3RMSNorm	256	256
`mlp.gate_proj.weight`	Linear	15360×3840	58,982,400
`mlp.up_proj.weight`	Linear	15360×3840	58,982,400
`mlp.down_proj.weight`	Linear	3840×15360	58,982,400
`input_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`post_attention_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`pre_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`post_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	3,840
Per block			**224,148,992**
× 40 blocks			**8,965,959,680**

Table 4: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 5, 11, …, 47 (every 6th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×3840	15,728,640
`self_attn.k_proj.weight`	Linear	2048×3840	7,864,320
`self_attn.v_proj.weight`	Linear	2048×3840	7,864,320
`self_attn.o_proj.weight`	Linear	3840×4096	15,728,640
`self_attn.q_norm.weight`	Gemma3RMSNorm	256	256
`self_attn.k_norm.weight`	Gemma3RMSNorm	256	256
`mlp.gate_proj.weight`	Linear	15360×3840	58,982,400
`mlp.up_proj.weight`	Linear	15360×3840	58,982,400
`mlp.down_proj.weight`	Linear	3840×15360	58,982,400
`input_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`post_attention_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`pre_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	3,840
`post_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	3,840
Per block			**224,148,992**
× 8 blocks			**1,793,191,936**

4.3 Vision encoder: `model.vision_tower.encoder.layers`

Table 5: $\mathcal{B}_V$: Vision encoder block — layers 0–26; weight names are relative to `model.vision_tower.encoder.layers.`ℓ

Weight (relative)	Type	Shape	Params
`layer_norm1.weight`	LayerNorm	1152	1,152
`layer_norm1.bias`	LayerNorm	1152	1,152
`self_attn.k_proj.weight`	Linear	1152×1152	1,327,104
`self_attn.k_proj.bias`	Linear	1152	1,152
`self_attn.v_proj.weight`	Linear	1152×1152	1,327,104
`self_attn.v_proj.bias`	Linear	1152	1,152
`self_attn.q_proj.weight`	Linear	1152×1152	1,327,104
`self_attn.q_proj.bias`	Linear	1152	1,152
`self_attn.out_proj.weight`	Linear	1152×1152	1,327,104
`self_attn.out_proj.bias`	Linear	1152	1,152
`layer_norm2.weight`	LayerNorm	1152	1,152
`layer_norm2.bias`	LayerNorm	1152	1,152
`mlp.fc1.weight`	Linear	4304×1152	4,958,208
`mlp.fc1.bias`	Linear	4304	4,304
`mlp.fc2.weight`	Linear	1152×4304	4,958,208
`mlp.fc2.bias`	Linear	1152	1,152
Per block			**15,239,504**
× 27 blocks			**411,466,608**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	RMSNorm applied to the attention output before it is added to the residual stream (Gemma "sandwich" norm).

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Layer (appears as)	What it does
Pre-MLP norm (pre_feedforward_layernorm) `layers.pre_feedforward_layernorm`	RMSNorm on the residual stream before the MLP (Gemma).
Post-MLP norm (post_feedforward_layernorm) `layers.post_feedforward_layernorm`	RMSNorm on the MLP output before it is added back (Gemma "sandwich" norm).
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.
VISION ENCODER	
Patch embedding (Conv2d) `embeddings.patch_embedding`	A strided convolution that cuts the image into square patches (14×14 pixels) and maps each patch to one vector.
Position embedding `embeddings.position_embedding`	One learned vector per patch position, added so the encoder knows where each patch is.
Pre-attention LayerNorm `layers.layer_norm1`	LayerNorm (mean/variance normalisation with scale and bias) before self-attention.
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Attention output projection `self_attn.out_proj`	Mixes the concatenated head outputs back to the model width.
Pre-MLP LayerNorm `layers.layer_norm2`	LayerNorm before the MLP.
MLP up-projection (fc1) `mlp.fc1`	Expands each vector to the wider MLP size before the GELU non-linearity.
MLP down-projection (fc2) `mlp.fc2`	Shrinks back to the model width.
Final vision LayerNorm `vision_tower.post_layernorm`	Normalises the vision encoder's output patches.
MULTIMODAL PROJECTOR	
Image-to-text projection `multi_modal_projector`	Matrix mapping each pooled image feature (vision width) into the language model's hidden size.
Projector norm `multi_modal_projector.mm_soft_emb_norm`	RMSNorm on image features after average-pooling the 4096 patches down to 256 image tokens.

6 Parameter Count

Table 7: Parameters by component

Component	Parameters	% of total
MLP / MoE	8,493,465,600	69.7%
Attention / token mixer	2,264,948,736	18.6%
Token embedding	1,006,878,720	8.3%
Vision encoder	416,866,032	3.4%
Multimodal projector	4,424,832	0.0%
Norms	741,120	0.0%
Total	**12,187,325,040**	**100.0%**

Table 8: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{S}}$ Sliding-window attention block	40	224,148,992	8,965,959,680
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	8	224,148,992	1,793,191,936
$\mathcal{B}_{\mathrm{V}}$ Vision encoder block	27	15,239,504	411,466,608
Modules outside stacks	1	–	1,016,706,816
Total			**12,187,325,040**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	262208 × 3840	bfloat16	1,006,878,720
`model`	Gemma3Model	–	–	–
`model.language_model`	Gemma3TextModel	–	–	–
`model.language_model.embed_tokens.weight`	Gemma3TextScaled WordEmbedding	262208 × 3840	bfloat16	1,006,878,720
`model.language_model.layers.0`	Gemma3DecoderLayer	–	–	–
`model.language_model.layers.0. input_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.0.mlp`	Gemma3MLP	–	–	–
`model.language_model.layers.0.mlp.down_proj. weight`	Linear	3840 × 15360	bfloat16	58,982,400
`model.language_model.layers.0.mlp.gate_proj. weight`	Linear	15360 × 3840	bfloat16	58,982,400
`model.language_model.layers.0.mlp.up_proj. weight`	Linear	15360 × 3840	bfloat16	58,982,400
`model.language_model.layers.0. post_attention_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.0. post_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.0. pre_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.0.self_attn`	Gemma3Attention	–	–	–
`model.language_model.layers.0.self_attn. k_norm.weight`	Gemma3RMSNorm	256	bfloat16	256
`model.language_model.layers.0.self_attn. k_proj.weight`	Linear	2048 × 3840	bfloat16	7,864,320
`model.language_model.layers.0.self_attn. o_proj.weight`	Linear	3840 × 4096	bfloat16	15,728,640
`model.language_model.layers.0.self_attn. q_norm.weight`	Gemma3RMSNorm	256	bfloat16	256
`model.language_model.layers.0.self_attn. q_proj.weight`	Linear	4096 × 3840	bfloat16	15,728,640
`model.language_model.layers.0.self_attn. v_proj.weight`	Linear	2048 × 3840	bfloat16	7,864,320
`model.language_model.layers.1`	Gemma3DecoderLayer	–	–	–
`model.language_model.layers.1. input_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.1.mlp`	Gemma3MLP	–	–	–
`model.language_model.layers.1.mlp.down_proj. weight`	Linear	3840 × 15360	bfloat16	58,982,400
`model.language_model.layers.1.mlp.gate_proj. weight`	Linear	15360 × 3840	bfloat16	58,982,400
`model.language_model.layers.1.mlp.up_proj. weight`	Linear	15360 × 3840	bfloat16	58,982,400
`model.language_model.layers.1. post_attention_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.1. post_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.1. pre_feedforward_layernorm.weight`	Gemma3RMSNorm	3840	bfloat16	3,840
`model.language_model.layers.1.self_attn`	Gemma3Attention	–	–	–
`model.language_model.layers.1.self_attn. k_norm.weight`	Gemma3RMSNorm	256	bfloat16	256
`model.language_model.layers.1.self_attn. k_proj.weight`	Linear	2048 × 3840	bfloat16	7,864,320

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.1.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.1.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.1.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.1.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.2	Gemma3DecoderLayer	–	–	–
model.language_model.layers.2.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.2.mlp	Gemma3MLP	–	–	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.2.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.2.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.2.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.2.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.2.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.2.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.2.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.2.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.2.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.2.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.2.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.2.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.3	Gemma3DecoderLayer	–	–	–
model.language_model.layers.3.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.3.mlp	Gemma3MLP	–	–	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.3.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.3.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.3.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.3.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.3.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.3.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.3.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.3.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.3.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.3.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.4	Gemma3DecoderLayer	–	–	–
model.language_model.layers.4.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.4.mlp	Gemma3MLP	–	–	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.4.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.4.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.4.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.4.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.4.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.4.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.4.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.4.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.4.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.4.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.4.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.4.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.5	Gemma3DecoderLayer	–	–	–
model.language_model.layers.5.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.5.mlp	Gemma3MLP	–	–	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.5.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.5.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.5.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.5.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.5.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.5.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.5.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.5.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.5.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.5.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.5.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.5.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.6	Gemma3DecoderLayer	–	–	–
model.language_model.layers.6.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.6.mlp	Gemma3MLP	–	–	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.6.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.6.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.6.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.6.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.6.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.6.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.6.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.6.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.6.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.6.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.6.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.6.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.7	Gemma3DecoderLayer	–	–	–
model.language_model.layers.7.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.7.mlp	Gemma3MLP	–	–	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.7.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.7.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.7.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.7.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.7.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.7.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.7.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.7.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.7.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.7.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.8	Gemma3DecoderLayer	–	–	–
model.language_model.layers.8.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.8.mlp	Gemma3MLP	–	–	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.8.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.8.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.8.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.8.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.8.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.8.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.8.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.8.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.8.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.8.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.8.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.8.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.9	Gemma3DecoderLayer	–	–	–
model.language_model.layers.9.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.9.mlp	Gemma3MLP	–	–	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.9.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.9.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.9.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.9.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.9.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.9.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.9.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.9.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.9.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.9.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.9.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.9.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.10	Gemma3DecoderLayer	–	–	–
model.language_model.layers.10.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.10.mlp	Gemma3MLP	–	–	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.10.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.10.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.10.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.10.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.10.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.10.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.10.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.10.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.10.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.10.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.10.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.10.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.11	Gemma3DecoderLayer	–	–	–
model.language_model.layers.11.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.11.mlp	Gemma3MLP	–	–	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.11.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.11.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.11.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.11.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.11.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.11.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.11.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.11.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.11.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.11.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.11.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.12	Gemma3DecoderLayer	–	–	–
model.language_model.layers.12.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.12.mlp	Gemma3MLP	–	–	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.12.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.12.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.12.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.12.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.12.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.12.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.12.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.12.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.12.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.12.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.12.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.12.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.13	Gemma3DecoderLayer	–	–	–
model.language_model.layers.13.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.13.mlp	Gemma3MLP	–	–	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.13.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.13.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.13.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.13.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.13.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.13.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.13.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.13.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.13.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.13.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.13.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.13.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.14	Gemma3DecoderLayer	–	–	–
model.language_model.layers.14.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.14.mlp	Gemma3MLP	–	–	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.14.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.14.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.14.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.14.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.14.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.14.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.14.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.14.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.14.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.14.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.14.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.14.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.15	Gemma3DecoderLayer	–	–	–
model.language_model.layers.15.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.15.mlp	Gemma3MLP	–	–	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.15.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.15.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.15.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.15.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.15.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.15.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.15.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.15.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.15.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.15.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.16	Gemma3DecoderLayer	–	–	–
model.language_model.layers.16.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.16.mlp	Gemma3MLP	–	–	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.16.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.16.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.16.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.16.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.16.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.16.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.16.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.16.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.16.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.16.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.16.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.16.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.17	Gemma3DecoderLayer	–	–	–
model.language_model.layers.17.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.17.mlp	Gemma3MLP	–	–	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.17.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.17.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.17.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.17.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.17.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.17.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.17.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.17.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.17.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.17.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.17.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.17.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.18	Gemma3DecoderLayer	–	–	–
model.language_model.layers.18.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.18.mlp	Gemma3MLP	–	–	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.18.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.18.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.18.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.18.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.18.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.18.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.18.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.18.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.18.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.18.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.18.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.18.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.19	Gemma3DecoderLayer	–	–	–
model.language_model.layers.19.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.19.mlp	Gemma3MLP	–	–	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.19.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.19.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.19.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.19.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.19.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.19.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.19.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.19.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.19.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.19.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.20	Gemma3DecoderLayer	–	–	–
model.language_model.layers.20.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.20.mlp	Gemma3MLP	–	–	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.20.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.20.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.20.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.20.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.20.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.20.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.20.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.20.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.20.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.20.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.20.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.20.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.21	Gemma3DecoderLayer	–	–	–
model.language_model.layers.21.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.21.mlp	Gemma3MLP	–	–	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.21.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.21.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.21.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.21.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.21.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.21.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.21.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.21.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.21.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.21.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.21.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.21.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.22	Gemma3DecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.22.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.22.mlp	Gemma3MLP	–	–	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.22.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.22.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.22.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.22.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.22.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.22.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.22.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.22.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.22.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.22.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.22.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.22.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.23	Gemma3DecoderLayer	–	–	–
model.language_model.layers.23.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.23.mlp	Gemma3MLP	–	–	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.23.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.23.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.23.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.23.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.23.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.23.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.23.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.23.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.23.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.23.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.24	Gemma3DecoderLayer	–	–	–
model.language_model.layers.24.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.24.mlp	Gemma3MLP	–	–	–
model.language_model.layers.24.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.24.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.24.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.24.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.24.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.24.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.24.self_attn	Gemma3Attention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.24.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.24.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.24.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.24.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.24.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.24.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.25	Gemma3DecoderLayer	–	–	–
model.language_model.layers.25.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.25.mlp	Gemma3MLP	–	–	–
model.language_model.layers.25.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.25.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.25.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.25.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.25.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.25.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.25.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.25.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.25.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.25.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.25.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.25.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.25.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.26	Gemma3DecoderLayer	–	–	–
model.language_model.layers.26.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.26.mlp	Gemma3MLP	–	–	–
model.language_model.layers.26.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.26.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.26.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.26.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.26.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.26.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.26.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.26.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.26.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.26.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.26.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.26.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.26.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.27	Gemma3DecoderLayer	–	–	–
model.language_model.layers.27.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.27.mlp	Gemma3MLP	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.27.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.27.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.27.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.27.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.27.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.27.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.27.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.27.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.27.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.27.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.27.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.27.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.28	Gemma3DecoderLayer	–	–	–
model.language_model.layers.28.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.28.mlp	Gemma3MLP	–	–	–
model.language_model.layers.28.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.28.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.28.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.28.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.28.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.28.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.28.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.28.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.28.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.28.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.28.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.28.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.28.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.29	Gemma3DecoderLayer	–	–	–
model.language_model.layers.29.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.29.mlp	Gemma3MLP	–	–	–
model.language_model.layers.29.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.29.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.29.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.29.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.29.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.29.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.29.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.29.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.29.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.29.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.29.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.29.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.29.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.30	Gemma3DecoderLayer	–	–	–
model.language_model.layers.30.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.30.mlp	Gemma3MLP	–	–	–
model.language_model.layers.30.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.30.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.30.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.30.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.30.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.30.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.30.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.30.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.30.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.30.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.30.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.30.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.30.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.31	Gemma3DecoderLayer	–	–	–
model.language_model.layers.31.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.31.mlp	Gemma3MLP	–	–	–
model.language_model.layers.31.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.31.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.31.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.31.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.31.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.31.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.31.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.31.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.31.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.31.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.31.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.31.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.32	Gemma3DecoderLayer	–	–	–
model.language_model.layers.32.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.32.mlp	Gemma3MLP	–	–	–
model.language_model.layers.32.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.32.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.32.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.32.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.32.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.32.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.32.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.32.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.32.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.32.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.32.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.32.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.32.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.33	Gemma3DecoderLayer	–	–	–
model.language_model.layers.33.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.33.mlp	Gemma3MLP	–	–	–
model.language_model.layers.33.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.33.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.33.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.33.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.33.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.33.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.33.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.33.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.33.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.33.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.33.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.33.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.33.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.34	Gemma3DecoderLayer	–	–	–
model.language_model.layers.34.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.34.mlp	Gemma3MLP	–	–	–
model.language_model.layers.34.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.34.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.34.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.34.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.34.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.34.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.34.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.34.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.34.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.34.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.34.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.34.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.34.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.35	Gemma3DecoderLayer	–	–	–
model.language_model.layers.35.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.35.mlp	Gemma3MLP	–	–	–
model.language_model.layers.35.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.35.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.35.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.35.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.35.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.35.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.35.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.35.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.35.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.35.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.35.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.35.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.35.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.36	Gemma3DecoderLayer	–	–	–
model.language_model.layers.36.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.36.mlp	Gemma3MLP	–	–	–
model.language_model.layers.36.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.36.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.36.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.36.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.36.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.36.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.36.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.36.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.36.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.36.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.36.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.36.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.36.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.37	Gemma3DecoderLayer	–	–	–
model.language_model.layers.37.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.37.mlp	Gemma3MLP	–	–	–
model.language_model.layers.37.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.37.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.37.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.37.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.37.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.37.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.37.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.37.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.37.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.37.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.37.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.37.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.37.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.38	Gemma3DecoderLayer	–	–	–
model.language_model.layers.38.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.38.mlp	Gemma3MLP	–	–	–
model.language_model.layers.38.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.38.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.38.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.38.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.38.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.38.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.38.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.38.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.38.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.38.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.38.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.38.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.38.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.39	Gemma3DecoderLayer	–	–	–
model.language_model.layers.39.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.39.mlp	Gemma3MLP	–	–	–
model.language_model.layers.39.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.39.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.39.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.39.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.39.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.39.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.39.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.39.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.39.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.39.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.39.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.39.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.39.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.40	Gemma3DecoderLayer	–	–	–
model.language_model.layers.40.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.40.mlp	Gemma3MLP	–	–	–
model.language_model.layers.40.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.40.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.40.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.40.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.40.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.40.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.40.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.40.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.40.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.40.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.40.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.40.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.40.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.41	Gemma3DecoderLayer	–	–	–
model.language_model.layers.41.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.41.mlp	Gemma3MLP	–	–	–
model.language_model.layers.41.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.41.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.41.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.41.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.41.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.41.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.41.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.41.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.41.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.41.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.41.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.41.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.41.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.42	Gemma3DecoderLayer	–	–	–
model.language_model.layers.42.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.42.mlp	Gemma3MLP	–	–	–
model.language_model.layers.42.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.42.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.42.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.42.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.42.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.42.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.42.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.42.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.42.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.42.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.42.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.42.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.42.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.43	Gemma3DecoderLayer	–	–	–
model.language_model.layers.43.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.43.mlp	Gemma3MLP	–	–	–
model.language_model.layers.43.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.43.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.43.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.43.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.43.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.43.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.43.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.43.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.43.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.43.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.43.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.43.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.43.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.44	Gemma3DecoderLayer	–	–	–
model.language_model.layers.44.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.44.mlp	Gemma3MLP	–	–	–
model.language_model.layers.44.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.44.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.44.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.44.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.44.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.44.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.44.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.44.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.44.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.44.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.44.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.44.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.44.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.45	Gemma3DecoderLayer	–	–	–
model.language_model.layers.45.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.45.mlp	Gemma3MLP	–	–	–
model.language_model.layers.45.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.45.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.45.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.45.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.45.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.45.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.45.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.45.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.45.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.45.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.45.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.45.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.45.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.46	Gemma3DecoderLayer	–	–	–
model.language_model.layers.46.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.46.mlp	Gemma3MLP	–	–	–
model.language_model.layers.46.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.46.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.46.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.46.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.46.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.46.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.46.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.46.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.46.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.46.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.46.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.46.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.46.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.47	Gemma3DecoderLayer	–	–	–
model.language_model.layers.47.input_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.47.mlp	Gemma3MLP	–	–	–
model.language_model.layers.47.mlp.down_proj.weight	Linear	3840 × 15360	bfloat16	58,982,400
model.language_model.layers.47.mlp.gate_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.47.mlp.up_proj.weight	Linear	15360 × 3840	bfloat16	58,982,400
model.language_model.layers.47.post_attention_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.47.post_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.47.pre_feedforward_layernorm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.language_model.layers.47.self_attn	Gemma3Attention	–	–	–
model.language_model.layers.47.self_attn.k_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.47.self_attn.k_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.layers.47.self_attn.o_proj.weight	Linear	3840 × 4096	bfloat16	15,728,640
model.language_model.layers.47.self_attn.q_norm.weight	Gemma3RMSNorm	256	bfloat16	256
model.language_model.layers.47.self_attn.q_proj.weight	Linear	4096 × 3840	bfloat16	15,728,640
model.language_model.layers.47.self_attn.v_proj.weight	Linear	2048 × 3840	bfloat16	7,864,320
model.language_model.norm.weight	Gemma3RMSNorm	3840	bfloat16	3,840
model.multi_modal_projector.mm_input_projection_weight	Gemma3MultiModalProjector	1152 × 3840	bfloat16	4,423,680
model.multi_modal_projector	Gemma3MultiModalProjector	–	–	–
model.multi_modal_projector.mm_soft_emb_norm.weight	Gemma3RMSNorm	1152	bfloat16	1,152
model.vision_tower	SiglipVisionModel	–	–	–
model.vision_tower.embeddings	SiglipVisionEmbeddings	–	–	–
model.vision_tower.embeddings.patch_embedding.bias	Conv2d	1152	bfloat16	1,152
model.vision_tower.embeddings.patch_embedding.weight	Conv2d	1152 × 3 × 14 × 14	bfloat16	677,376
model.vision_tower.embeddings.position_embedding.weight	Embedding	4096 × 1152	bfloat16	4,718,592
model.vision_tower.encoder	SiglipEncoder	–	–	–
model.vision_tower.encoder.layers.0	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.0.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.0.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.0.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.0.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.0.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.0.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.0.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.0.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.0.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.0.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.1	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.1.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.1.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.1.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.1.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.1.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.1.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.1.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.1.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.1.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.1.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.1.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.2	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.2.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.2.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.2.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.2.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.2.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.2.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.2.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.2.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.2.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.2.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.3	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.3.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.3.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.3.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.3.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.3.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.3.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.3.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.3.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.3.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.3.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.3.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.4	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.4.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.4.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.4.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.4.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.4.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.4.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.4.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.4.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.4.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.4.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.5	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.5.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.5.mlp.fc1.bias	Linear	4304	bfloat16	4,304

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.5.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.5.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.5.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.5.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.5.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.5.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.5.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.5.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.6	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.6.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.6.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.6.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.6.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.6.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.6.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.6.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.6.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.6.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.6.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.7	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.7.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.7.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.7.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.7.mlp.fc2.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.7.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.7.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.7.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.7.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.7.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.7.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.7.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.8	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.8.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.8.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.8.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.8.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.8.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.8.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.8.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.8.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.8.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.8.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.9	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.9.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.9.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.9.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.9.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.9.self_attn	SiglipAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.9.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.9.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.9.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.9.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.9.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.10	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.10.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.10.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.10.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.10.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.10.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.10.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.10.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.10.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.10.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.10.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.11	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.11.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.11.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.11.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.11.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.11.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.11.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.11.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.11.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.11.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.11.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.11.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.12	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.12.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.12.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.12.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.12.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.12.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.12.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.12.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.12.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.12.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.12.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.13	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.13.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.13.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.13.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.13.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.13.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.13.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.13.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.13.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.13.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.13.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.14	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.14.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.14.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.14.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.14.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.14.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.14.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.14.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.14.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.14.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.14.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.15	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.15.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.15.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.15.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.15.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.15.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.15.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.15.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.15.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.15.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.15.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.15.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.16	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.16.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.16.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.16.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.16.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.16.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.16.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.16.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.16.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.16.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.16.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.17	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.17.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.17.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.17.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.17.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.17.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.17.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.17.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.17.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.17.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.17.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.18	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.18.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.18.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.18.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.18.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.18.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.18.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.18.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.18.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.18.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.18.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.19	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.19.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.19.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.19.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.19.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.19.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.19.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.19.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.19.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.19.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.19.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.19.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.20	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.20.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.20.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.20.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.20.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.20.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.20.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.20.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.20.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.20.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.20.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.21	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.21.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.21.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.21.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.21.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.21.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.21.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.21.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.21.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.21.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.21.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.22	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.22.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.22.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.22.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.22.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.22.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.22.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.22.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.22.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.22.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.22.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.23	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.23.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.23.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.23.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.23.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.23.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.23.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.23.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.23.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.23.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.23.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.23.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.24	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.24.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.24.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.24.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.24.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.24.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.24.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.24.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.24.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.24.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.24.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.25	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.25.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.25.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.25.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.25.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.25.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.25.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.25.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.25.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.25.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.25.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104

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Layer path	Type	Shape	Dtype	Params
model.vision_tower.encoder.layers.26	SiglipEncoderLayer	–	–	–
model.vision_tower.encoder.layers.26.layer_norm1.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.layer_norm1.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.layer_norm2.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.layer_norm2.weight	LayerNorm	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.mlp	SiglipMLP	–	–	–
model.vision_tower.encoder.layers.26.mlp.fc1.bias	Linear	4304	bfloat16	4,304
model.vision_tower.encoder.layers.26.mlp.fc1.weight	Linear	4304 × 1152	bfloat16	4,958,208
model.vision_tower.encoder.layers.26.mlp.fc2.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.mlp.fc2.weight	Linear	1152 × 4304	bfloat16	4,958,208
model.vision_tower.encoder.layers.26.self_attn	SiglipAttention	–	–	–
model.vision_tower.encoder.layers.26.self_attn.k_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.self_attn.k_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.26.self_attn.out_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.self_attn.out_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.26.self_attn.q_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.self_attn.q_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.encoder.layers.26.self_attn.v_proj.bias	Linear	1152	bfloat16	1,152
model.vision_tower.encoder.layers.26.self_attn.v_proj.weight	Linear	1152 × 1152	bfloat16	1,327,104
model.vision_tower.post_layernorm.bias	LayerNorm	1152	bfloat16	1,152
model.vision_tower.post_layernorm.weight	LayerNorm	1152	bfloat16	1,152

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.language_model.embed_tokens	1,006,878,720	1,006,878,720
model.language_model.layers.0.input_layernorm	3,840	1,006,882,560
model.language_model.layers.0.mlp.down_proj	58,982,400	1,065,864,960
model.language_model.layers.0.mlp.gate_proj	58,982,400	1,124,847,360
model.language_model.layers.0.mlp.up_proj	58,982,400	1,183,829,760
model.language_model.layers.0.post_attention_layernorm	3,840	1,183,833,600
model.language_model.layers.0.post_feedforward_layernorm	3,840	1,183,837,440
model.language_model.layers.0.pre_feedforward_layernorm	3,840	1,183,841,280
model.language_model.layers.0.self_attn.k_norm	256	1,183,841,536
model.language_model.layers.0.self_attn.k_proj	7,864,320	1,191,705,856
model.language_model.layers.0.self_attn.o_proj	15,728,640	1,207,434,496
model.language_model.layers.0.self_attn.q_norm	256	1,207,434,752
model.language_model.layers.0.self_attn.q_proj	15,728,640	1,223,163,392
model.language_model.layers.0.self_attn.v_proj	7,864,320	1,231,027,712
model.language_model.layers.1.input_layernorm	3,840	1,231,031,552
model.language_model.layers.1.mlp.down_proj	58,982,400	1,290,013,952
model.language_model.layers.1.mlp.gate_proj	58,982,400	1,348,996,352
model.language_model.layers.1.mlp.up_proj	58,982,400	1,407,978,752
model.language_model.layers.1.post_attention_layernorm	3,840	1,407,982,592

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Module	Params	Running total
model.language_model.layers.1. post_feedforward_layernorm	3,840	1,407,986,432
model.language_model.layers.1. pre_feedforward_layernorm	3,840	1,407,990,272
model.language_model.layers.1.self_attn.k_norm	256	1,407,990,528
model.language_model.layers.1.self_attn.k_proj	7,864,320	1,415,854,848
model.language_model.layers.1.self_attn.o_proj	15,728,640	1,431,583,488
model.language_model.layers.1.self_attn.q_norm	256	1,431,583,744
model.language_model.layers.1.self_attn.q_proj	15,728,640	1,447,312,384
model.language_model.layers.1.self_attn.v_proj	7,864,320	1,455,176,704
model.language_model.layers.2.input_layernorm	3,840	1,455,180,544
model.language_model.layers.2.mlp.down_proj	58,982,400	1,514,162,944
model.language_model.layers.2.mlp.gate_proj	58,982,400	1,573,145,344
model.language_model.layers.2.mlp.up_proj	58,982,400	1,632,127,744
model.language_model.layers.2. post_attention_layernorm	3,840	1,632,131,584
model.language_model.layers.2. post_feedforward_layernorm	3,840	1,632,135,424
model.language_model.layers.2. pre_feedforward_layernorm	3,840	1,632,139,264
model.language_model.layers.2.self_attn.k_norm	256	1,632,139,520
model.language_model.layers.2.self_attn.k_proj	7,864,320	1,640,003,840
model.language_model.layers.2.self_attn.o_proj	15,728,640	1,655,732,480
model.language_model.layers.2.self_attn.q_norm	256	1,655,732,736
model.language_model.layers.2.self_attn.q_proj	15,728,640	1,671,461,376
model.language_model.layers.2.self_attn.v_proj	7,864,320	1,679,325,696
model.language_model.layers.3.input_layernorm	3,840	1,679,329,536
model.language_model.layers.3.mlp.down_proj	58,982,400	1,738,311,936
model.language_model.layers.3.mlp.gate_proj	58,982,400	1,797,294,336
model.language_model.layers.3.mlp.up_proj	58,982,400	1,856,276,736
model.language_model.layers.3. post_attention_layernorm	3,840	1,856,280,576
model.language_model.layers.3. post_feedforward_layernorm	3,840	1,856,284,416
model.language_model.layers.3. pre_feedforward_layernorm	3,840	1,856,288,256
model.language_model.layers.3.self_attn.k_norm	256	1,856,288,512
model.language_model.layers.3.self_attn.k_proj	7,864,320	1,864,152,832
model.language_model.layers.3.self_attn.o_proj	15,728,640	1,879,881,472
model.language_model.layers.3.self_attn.q_norm	256	1,879,881,728
model.language_model.layers.3.self_attn.q_proj	15,728,640	1,895,610,368
model.language_model.layers.3.self_attn.v_proj	7,864,320	1,903,474,688
model.language_model.layers.4.input_layernorm	3,840	1,903,478,528
model.language_model.layers.4.mlp.down_proj	58,982,400	1,962,460,928
model.language_model.layers.4.mlp.gate_proj	58,982,400	2,021,443,328
model.language_model.layers.4.mlp.up_proj	58,982,400	2,080,425,728
model.language_model.layers.4. post_attention_layernorm	3,840	2,080,429,568
model.language_model.layers.4. post_feedforward_layernorm	3,840	2,080,433,408
model.language_model.layers.4. pre_feedforward_layernorm	3,840	2,080,437,248
model.language_model.layers.4.self_attn.k_norm	256	2,080,437,504
model.language_model.layers.4.self_attn.k_proj	7,864,320	2,088,301,824
model.language_model.layers.4.self_attn.o_proj	15,728,640	2,104,030,464
model.language_model.layers.4.self_attn.q_norm	256	2,104,030,720
model.language_model.layers.4.self_attn.q_proj	15,728,640	2,119,759,360
model.language_model.layers.4.self_attn.v_proj	7,864,320	2,127,623,680

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Module	Params	Running total
model.language_model.layers.5.input_layernorm	3,840	2,127,627,520
model.language_model.layers.5.mlp.down_proj	58,982,400	2,186,609,920
model.language_model.layers.5.mlp.gate_proj	58,982,400	2,245,592,320
model.language_model.layers.5.mlp.up_proj	58,982,400	2,304,574,720
model.language_model.layers.5.post_attention_layernorm	3,840	2,304,578,560
model.language_model.layers.5.post_feedforward_layernorm	3,840	2,304,582,400
model.language_model.layers.5.pre_feedforward_layernorm	3,840	2,304,586,240
model.language_model.layers.5.self_attn.k_norm	256	2,304,586,496
model.language_model.layers.5.self_attn.k_proj	7,864,320	2,312,450,816
model.language_model.layers.5.self_attn.o_proj	15,728,640	2,328,179,456
model.language_model.layers.5.self_attn.q_norm	256	2,328,179,712
model.language_model.layers.5.self_attn.q_proj	15,728,640	2,343,908,352
model.language_model.layers.5.self_attn.v_proj	7,864,320	2,351,772,672
model.language_model.layers.6.input_layernorm	3,840	2,351,776,512
model.language_model.layers.6.mlp.down_proj	58,982,400	2,410,758,912
model.language_model.layers.6.mlp.gate_proj	58,982,400	2,469,741,312
model.language_model.layers.6.mlp.up_proj	58,982,400	2,528,723,712
model.language_model.layers.6.post_attention_layernorm	3,840	2,528,727,552
model.language_model.layers.6.post_feedforward_layernorm	3,840	2,528,731,392
model.language_model.layers.6.pre_feedforward_layernorm	3,840	2,528,735,232
model.language_model.layers.6.self_attn.k_norm	256	2,528,735,488
model.language_model.layers.6.self_attn.k_proj	7,864,320	2,536,599,808
model.language_model.layers.6.self_attn.o_proj	15,728,640	2,552,328,448
model.language_model.layers.6.self_attn.q_norm	256	2,552,328,704
model.language_model.layers.6.self_attn.q_proj	15,728,640	2,568,057,344
model.language_model.layers.6.self_attn.v_proj	7,864,320	2,575,921,664
model.language_model.layers.7.input_layernorm	3,840	2,575,925,504
model.language_model.layers.7.mlp.down_proj	58,982,400	2,634,907,904
model.language_model.layers.7.mlp.gate_proj	58,982,400	2,693,890,304
model.language_model.layers.7.mlp.up_proj	58,982,400	2,752,872,704
model.language_model.layers.7.post_attention_layernorm	3,840	2,752,876,544
model.language_model.layers.7.post_feedforward_layernorm	3,840	2,752,880,384
model.language_model.layers.7.pre_feedforward_layernorm	3,840	2,752,884,224
model.language_model.layers.7.self_attn.k_norm	256	2,752,884,480
model.language_model.layers.7.self_attn.k_proj	7,864,320	2,760,748,800
model.language_model.layers.7.self_attn.o_proj	15,728,640	2,776,477,440
model.language_model.layers.7.self_attn.q_norm	256	2,776,477,696
model.language_model.layers.7.self_attn.q_proj	15,728,640	2,792,206,336
model.language_model.layers.7.self_attn.v_proj	7,864,320	2,800,070,656
model.language_model.layers.8.input_layernorm	3,840	2,800,074,496
model.language_model.layers.8.mlp.down_proj	58,982,400	2,859,056,896
model.language_model.layers.8.mlp.gate_proj	58,982,400	2,918,039,296
model.language_model.layers.8.mlp.up_proj	58,982,400	2,977,021,696
model.language_model.layers.8.post_attention_layernorm	3,840	2,977,025,536
model.language_model.layers.8.post_feedforward_layernorm	3,840	2,977,029,376
model.language_model.layers.8.pre_feedforward_layernorm	3,840	2,977,033,216

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Module	Params	Running total
model.language_model.layers.8.self_attn.k_norm	256	2,977,033,472
model.language_model.layers.8.self_attn.k_proj	7,864,320	2,984,897,792
model.language_model.layers.8.self_attn.o_proj	15,728,640	3,000,626,432
model.language_model.layers.8.self_attn.q_norm	256	3,000,626,688
model.language_model.layers.8.self_attn.q_proj	15,728,640	3,016,355,328
model.language_model.layers.8.self_attn.v_proj	7,864,320	3,024,219,648
model.language_model.layers.9.input_layernorm	3,840	3,024,223,488
model.language_model.layers.9.mlp.down_proj	58,982,400	3,083,205,888
model.language_model.layers.9.mlp.gate_proj	58,982,400	3,142,188,288
model.language_model.layers.9.mlp.up_proj	58,982,400	3,201,170,688
model.language_model.layers.9. post_attention_layernorm	3,840	3,201,174,528
model.language_model.layers.9. post_feedforward_layernorm	3,840	3,201,178,368
model.language_model.layers.9. pre_feedforward_layernorm	3,840	3,201,182,208
model.language_model.layers.9.self_attn.k_norm	256	3,201,182,464
model.language_model.layers.9.self_attn.k_proj	7,864,320	3,209,046,784
model.language_model.layers.9.self_attn.o_proj	15,728,640	3,224,775,424
model.language_model.layers.9.self_attn.q_norm	256	3,224,775,680
model.language_model.layers.9.self_attn.q_proj	15,728,640	3,240,504,320
model.language_model.layers.9.self_attn.v_proj	7,864,320	3,248,368,640
model.language_model.layers.10.input_layernorm	3,840	3,248,372,480
model.language_model.layers.10.mlp.down_proj	58,982,400	3,307,354,880
model.language_model.layers.10.mlp.gate_proj	58,982,400	3,366,337,280
model.language_model.layers.10.mlp.up_proj	58,982,400	3,425,319,680
model.language_model.layers.10. post_attention_layernorm	3,840	3,425,323,520
model.language_model.layers.10. post_feedforward_layernorm	3,840	3,425,327,360
model.language_model.layers.10. pre_feedforward_layernorm	3,840	3,425,331,200
model.language_model.layers.10.self_attn.k_norm	256	3,425,331,456
model.language_model.layers.10.self_attn.k_proj	7,864,320	3,433,195,776
model.language_model.layers.10.self_attn.o_proj	15,728,640	3,448,924,416
model.language_model.layers.10.self_attn.q_norm	256	3,448,924,672
model.language_model.layers.10.self_attn.q_proj	15,728,640	3,464,653,312
model.language_model.layers.10.self_attn.v_proj	7,864,320	3,472,517,632
model.language_model.layers.11.input_layernorm	3,840	3,472,521,472
model.language_model.layers.11.mlp.down_proj	58,982,400	3,531,503,872
model.language_model.layers.11.mlp.gate_proj	58,982,400	3,590,486,272
model.language_model.layers.11.mlp.up_proj	58,982,400	3,649,468,672
model.language_model.layers.11. post_attention_layernorm	3,840	3,649,472,512
model.language_model.layers.11. post_feedforward_layernorm	3,840	3,649,476,352
model.language_model.layers.11. pre_feedforward_layernorm	3,840	3,649,480,192
model.language_model.layers.11.self_attn.k_norm	256	3,649,480,448
model.language_model.layers.11.self_attn.k_proj	7,864,320	3,657,344,768
model.language_model.layers.11.self_attn.o_proj	15,728,640	3,673,073,408
model.language_model.layers.11.self_attn.q_norm	256	3,673,073,664
model.language_model.layers.11.self_attn.q_proj	15,728,640	3,688,802,304
model.language_model.layers.11.self_attn.v_proj	7,864,320	3,696,666,624
model.language_model.layers.12.input_layernorm	3,840	3,696,670,464
model.language_model.layers.12.mlp.down_proj	58,982,400	3,755,652,864
model.language_model.layers.12.mlp.gate_proj	58,982,400	3,814,635,264
model.language_model.layers.12.mlp.up_proj	58,982,400	3,873,617,664

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Module	Params	Running total
model.language_model.layers.12.post_attention_layernorm	3,840	3,873,621,504
model.language_model.layers.12.post_feedforward_layernorm	3,840	3,873,625,344
model.language_model.layers.12.pre_feedforward_layernorm	3,840	3,873,629,184
model.language_model.layers.12.self_attn.k_norm	256	3,873,629,440
model.language_model.layers.12.self_attn.k_proj	7,864,320	3,881,493,760
model.language_model.layers.12.self_attn.o_proj	15,728,640	3,897,222,400
model.language_model.layers.12.self_attn.q_norm	256	3,897,222,656
model.language_model.layers.12.self_attn.q_proj	15,728,640	3,912,951,296
model.language_model.layers.12.self_attn.v_proj	7,864,320	3,920,815,616
model.language_model.layers.13.input_layernorm	3,840	3,920,819,456
model.language_model.layers.13.mlp.down_proj	58,982,400	3,979,801,856
model.language_model.layers.13.mlp.gate_proj	58,982,400	4,038,784,256
model.language_model.layers.13.mlp.up_proj	58,982,400	4,097,766,656
model.language_model.layers.13.post_attention_layernorm	3,840	4,097,770,496
model.language_model.layers.13.post_feedforward_layernorm	3,840	4,097,774,336
model.language_model.layers.13.pre_feedforward_layernorm	3,840	4,097,778,176
model.language_model.layers.13.self_attn.k_norm	256	4,097,778,432
model.language_model.layers.13.self_attn.k_proj	7,864,320	4,105,642,752
model.language_model.layers.13.self_attn.o_proj	15,728,640	4,121,371,392
model.language_model.layers.13.self_attn.q_norm	256	4,121,371,648
model.language_model.layers.13.self_attn.q_proj	15,728,640	4,137,100,288
model.language_model.layers.13.self_attn.v_proj	7,864,320	4,144,964,608
model.language_model.layers.14.input_layernorm	3,840	4,144,968,448
model.language_model.layers.14.mlp.down_proj	58,982,400	4,203,950,848
model.language_model.layers.14.mlp.gate_proj	58,982,400	4,262,933,248
model.language_model.layers.14.mlp.up_proj	58,982,400	4,321,915,648
model.language_model.layers.14.post_attention_layernorm	3,840	4,321,919,488
model.language_model.layers.14.post_feedforward_layernorm	3,840	4,321,923,328
model.language_model.layers.14.pre_feedforward_layernorm	3,840	4,321,927,168
model.language_model.layers.14.self_attn.k_norm	256	4,321,927,424
model.language_model.layers.14.self_attn.k_proj	7,864,320	4,329,791,744
model.language_model.layers.14.self_attn.o_proj	15,728,640	4,345,520,384
model.language_model.layers.14.self_attn.q_norm	256	4,345,520,640
model.language_model.layers.14.self_attn.q_proj	15,728,640	4,361,249,280
model.language_model.layers.14.self_attn.v_proj	7,864,320	4,369,113,600
model.language_model.layers.15.input_layernorm	3,840	4,369,117,440
model.language_model.layers.15.mlp.down_proj	58,982,400	4,428,099,840
model.language_model.layers.15.mlp.gate_proj	58,982,400	4,487,082,240
model.language_model.layers.15.mlp.up_proj	58,982,400	4,546,064,640
model.language_model.layers.15.post_attention_layernorm	3,840	4,546,068,480
model.language_model.layers.15.post_feedforward_layernorm	3,840	4,546,072,320
model.language_model.layers.15.pre_feedforward_layernorm	3,840	4,546,076,160
model.language_model.layers.15.self_attn.k_norm	256	4,546,076,416
model.language_model.layers.15.self_attn.k_proj	7,864,320	4,553,940,736
model.language_model.layers.15.self_attn.o_proj	15,728,640	4,569,669,376
model.language_model.layers.15.self_attn.q_norm	256	4,569,669,632

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Module	Params	Running total
model.language_model.layers.15.self_attn.q_proj	15,728,640	4,585,398,272
model.language_model.layers.15.self_attn.v_proj	7,864,320	4,593,262,592
model.language_model.layers.16.input_layernorm	3,840	4,593,266,432
model.language_model.layers.16.mlp.down_proj	58,982,400	4,652,248,832
model.language_model.layers.16.mlp.gate_proj	58,982,400	4,711,231,232
model.language_model.layers.16.mlp.up_proj	58,982,400	4,770,213,632
model.language_model.layers.16.post_attention_layernorm	3,840	4,770,217,472
model.language_model.layers.16.post_feedforward_layernorm	3,840	4,770,221,312
model.language_model.layers.16.pre_feedforward_layernorm	3,840	4,770,225,152
model.language_model.layers.16.self_attn.k_norm	256	4,770,225,408
model.language_model.layers.16.self_attn.k_proj	7,864,320	4,778,089,728
model.language_model.layers.16.self_attn.o_proj	15,728,640	4,793,818,368
model.language_model.layers.16.self_attn.q_norm	256	4,793,818,624
model.language_model.layers.16.self_attn.q_proj	15,728,640	4,809,547,264
model.language_model.layers.16.self_attn.v_proj	7,864,320	4,817,411,584
model.language_model.layers.17.input_layernorm	3,840	4,817,415,424
model.language_model.layers.17.mlp.down_proj	58,982,400	4,876,397,824
model.language_model.layers.17.mlp.gate_proj	58,982,400	4,935,380,224
model.language_model.layers.17.mlp.up_proj	58,982,400	4,994,362,624
model.language_model.layers.17.post_attention_layernorm	3,840	4,994,366,464
model.language_model.layers.17.post_feedforward_layernorm	3,840	4,994,370,304
model.language_model.layers.17.pre_feedforward_layernorm	3,840	4,994,374,144
model.language_model.layers.17.self_attn.k_norm	256	4,994,374,400
model.language_model.layers.17.self_attn.k_proj	7,864,320	5,002,238,720
model.language_model.layers.17.self_attn.o_proj	15,728,640	5,017,967,360
model.language_model.layers.17.self_attn.q_norm	256	5,017,967,616
model.language_model.layers.17.self_attn.q_proj	15,728,640	5,033,696,256
model.language_model.layers.17.self_attn.v_proj	7,864,320	5,041,560,576
model.language_model.layers.18.input_layernorm	3,840	5,041,564,416
model.language_model.layers.18.mlp.down_proj	58,982,400	5,100,546,816
model.language_model.layers.18.mlp.gate_proj	58,982,400	5,159,529,216
model.language_model.layers.18.mlp.up_proj	58,982,400	5,218,511,616
model.language_model.layers.18.post_attention_layernorm	3,840	5,218,515,456
model.language_model.layers.18.post_feedforward_layernorm	3,840	5,218,519,296
model.language_model.layers.18.pre_feedforward_layernorm	3,840	5,218,523,136
model.language_model.layers.18.self_attn.k_norm	256	5,218,523,392
model.language_model.layers.18.self_attn.k_proj	7,864,320	5,226,387,712
model.language_model.layers.18.self_attn.o_proj	15,728,640	5,242,116,352
model.language_model.layers.18.self_attn.q_norm	256	5,242,116,608
model.language_model.layers.18.self_attn.q_proj	15,728,640	5,257,845,248
model.language_model.layers.18.self_attn.v_proj	7,864,320	5,265,709,568
model.language_model.layers.19.input_layernorm	3,840	5,265,713,408
model.language_model.layers.19.mlp.down_proj	58,982,400	5,324,695,808
model.language_model.layers.19.mlp.gate_proj	58,982,400	5,383,678,208
model.language_model.layers.19.mlp.up_proj	58,982,400	5,442,660,608
model.language_model.layers.19.post_attention_layernorm	3,840	5,442,664,448
model.language_model.layers.19.post_feedforward_layernorm	3,840	5,442,668,288

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Module	Params	Running total
model.language_model.layers.19. pre_feedforward_layernorm	3,840	5,442,672,128
model.language_model.layers.19.self_attn.k_norm	256	5,442,672,384
model.language_model.layers.19.self_attn.k_proj	7,864,320	5,450,536,704
model.language_model.layers.19.self_attn.o_proj	15,728,640	5,466,265,344
model.language_model.layers.19.self_attn.q_norm	256	5,466,265,600
model.language_model.layers.19.self_attn.q_proj	15,728,640	5,481,994,240
model.language_model.layers.19.self_attn.v_proj	7,864,320	5,489,858,560
model.language_model.layers.20.input_layernorm	3,840	5,489,862,400
model.language_model.layers.20.mlp.down_proj	58,982,400	5,548,844,800
model.language_model.layers.20.mlp.gate_proj	58,982,400	5,607,827,200
model.language_model.layers.20.mlp.up_proj	58,982,400	5,666,809,600
model.language_model.layers.20. post_attention_layernorm	3,840	5,666,813,440
model.language_model.layers.20. post_feedforward_layernorm	3,840	5,666,817,280
model.language_model.layers.20. pre_feedforward_layernorm	3,840	5,666,821,120
model.language_model.layers.20.self_attn.k_norm	256	5,666,821,376
model.language_model.layers.20.self_attn.k_proj	7,864,320	5,674,685,696
model.language_model.layers.20.self_attn.o_proj	15,728,640	5,690,414,336
model.language_model.layers.20.self_attn.q_norm	256	5,690,414,592
model.language_model.layers.20.self_attn.q_proj	15,728,640	5,706,143,232
model.language_model.layers.20.self_attn.v_proj	7,864,320	5,714,007,552
model.language_model.layers.21.input_layernorm	3,840	5,714,011,392
model.language_model.layers.21.mlp.down_proj	58,982,400	5,772,993,792
model.language_model.layers.21.mlp.gate_proj	58,982,400	5,831,976,192
model.language_model.layers.21.mlp.up_proj	58,982,400	5,890,958,592
model.language_model.layers.21. post_attention_layernorm	3,840	5,890,962,432
model.language_model.layers.21. post_feedforward_layernorm	3,840	5,890,966,272
model.language_model.layers.21. pre_feedforward_layernorm	3,840	5,890,970,112
model.language_model.layers.21.self_attn.k_norm	256	5,890,970,368
model.language_model.layers.21.self_attn.k_proj	7,864,320	5,898,834,688
model.language_model.layers.21.self_attn.o_proj	15,728,640	5,914,563,328
model.language_model.layers.21.self_attn.q_norm	256	5,914,563,584
model.language_model.layers.21.self_attn.q_proj	15,728,640	5,930,292,224
model.language_model.layers.21.self_attn.v_proj	7,864,320	5,938,156,544
model.language_model.layers.22.input_layernorm	3,840	5,938,160,384
model.language_model.layers.22.mlp.down_proj	58,982,400	5,997,142,784
model.language_model.layers.22.mlp.gate_proj	58,982,400	6,056,125,184
model.language_model.layers.22.mlp.up_proj	58,982,400	6,115,107,584
model.language_model.layers.22. post_attention_layernorm	3,840	6,115,111,424
model.language_model.layers.22. post_feedforward_layernorm	3,840	6,115,115,264
model.language_model.layers.22. pre_feedforward_layernorm	3,840	6,115,119,104
model.language_model.layers.22.self_attn.k_norm	256	6,115,119,360
model.language_model.layers.22.self_attn.k_proj	7,864,320	6,122,983,680
model.language_model.layers.22.self_attn.o_proj	15,728,640	6,138,712,320
model.language_model.layers.22.self_attn.q_norm	256	6,138,712,576
model.language_model.layers.22.self_attn.q_proj	15,728,640	6,154,441,216
model.language_model.layers.22.self_attn.v_proj	7,864,320	6,162,305,536
model.language_model.layers.23.input_layernorm	3,840	6,162,309,376
model.language_model.layers.23.mlp.down_proj	58,982,400	6,221,291,776

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Module	Params	Running total
model.language_model.layers.23.mlp.gate_proj	58,982,400	6,280,274,176
model.language_model.layers.23.mlp.up_proj	58,982,400	6,339,256,576
model.language_model.layers.23. post_attention_layernorm	3,840	6,339,260,416
model.language_model.layers.23. post_feedforward_layernorm	3,840	6,339,264,256
model.language_model.layers.23. pre_feedforward_layernorm	3,840	6,339,268,096
model.language_model.layers.23.self_attn.k_norm	256	6,339,268,352
model.language_model.layers.23.self_attn.k_proj	7,864,320	6,347,132,672
model.language_model.layers.23.self_attn.o_proj	15,728,640	6,362,861,312
model.language_model.layers.23.self_attn.q_norm	256	6,362,861,568
model.language_model.layers.23.self_attn.q_proj	15,728,640	6,378,590,208
model.language_model.layers.23.self_attn.v_proj	7,864,320	6,386,454,528
model.language_model.layers.24.input_layernorm	3,840	6,386,458,368
model.language_model.layers.24.mlp.down_proj	58,982,400	6,445,440,768
model.language_model.layers.24.mlp.gate_proj	58,982,400	6,504,423,168
model.language_model.layers.24.mlp.up_proj	58,982,400	6,563,405,568
model.language_model.layers.24. post_attention_layernorm	3,840	6,563,409,408
model.language_model.layers.24. post_feedforward_layernorm	3,840	6,563,413,248
model.language_model.layers.24. pre_feedforward_layernorm	3,840	6,563,417,088
model.language_model.layers.24.self_attn.k_norm	256	6,563,417,344
model.language_model.layers.24.self_attn.k_proj	7,864,320	6,571,281,664
model.language_model.layers.24.self_attn.o_proj	15,728,640	6,587,010,304
model.language_model.layers.24.self_attn.q_norm	256	6,587,010,560
model.language_model.layers.24.self_attn.q_proj	15,728,640	6,602,739,200
model.language_model.layers.24.self_attn.v_proj	7,864,320	6,610,603,520
model.language_model.layers.25.input_layernorm	3,840	6,610,607,360
model.language_model.layers.25.mlp.down_proj	58,982,400	6,669,589,760
model.language_model.layers.25.mlp.gate_proj	58,982,400	6,728,572,160
model.language_model.layers.25.mlp.up_proj	58,982,400	6,787,554,560
model.language_model.layers.25. post_attention_layernorm	3,840	6,787,558,400
model.language_model.layers.25. post_feedforward_layernorm	3,840	6,787,562,240
model.language_model.layers.25. pre_feedforward_layernorm	3,840	6,787,566,080
model.language_model.layers.25.self_attn.k_norm	256	6,787,566,336
model.language_model.layers.25.self_attn.k_proj	7,864,320	6,795,430,656
model.language_model.layers.25.self_attn.o_proj	15,728,640	6,811,159,296
model.language_model.layers.25.self_attn.q_norm	256	6,811,159,552
model.language_model.layers.25.self_attn.q_proj	15,728,640	6,826,888,192
model.language_model.layers.25.self_attn.v_proj	7,864,320	6,834,752,512
model.language_model.layers.26.input_layernorm	3,840	6,834,756,352
model.language_model.layers.26.mlp.down_proj	58,982,400	6,893,738,752
model.language_model.layers.26.mlp.gate_proj	58,982,400	6,952,721,152
model.language_model.layers.26.mlp.up_proj	58,982,400	7,011,703,552
model.language_model.layers.26. post_attention_layernorm	3,840	7,011,707,392
model.language_model.layers.26. post_feedforward_layernorm	3,840	7,011,711,232
model.language_model.layers.26. pre_feedforward_layernorm	3,840	7,011,715,072
model.language_model.layers.26.self_attn.k_norm	256	7,011,715,328
model.language_model.layers.26.self_attn.k_proj	7,864,320	7,019,579,648

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Module	Params	Running total
model.language_model.layers.26.self_attn.o_proj	15,728,640	7,035,308,288
model.language_model.layers.26.self_attn.q_norm	256	7,035,308,544
model.language_model.layers.26.self_attn.q_proj	15,728,640	7,051,037,184
model.language_model.layers.26.self_attn.v_proj	7,864,320	7,058,901,504
model.language_model.layers.27.input_layernorm	3,840	7,058,905,344
model.language_model.layers.27.mlp.down_proj	58,982,400	7,117,887,744
model.language_model.layers.27.mlp.gate_proj	58,982,400	7,176,870,144
model.language_model.layers.27.mlp.up_proj	58,982,400	7,235,852,544
model.language_model.layers.27.post_attention_layernorm	3,840	7,235,856,384
model.language_model.layers.27.post_feedforward_layernorm	3,840	7,235,860,224
model.language_model.layers.27.pre_feedforward_layernorm	3,840	7,235,864,064
model.language_model.layers.27.self_attn.k_norm	256	7,235,864,320
model.language_model.layers.27.self_attn.k_proj	7,864,320	7,243,728,640
model.language_model.layers.27.self_attn.o_proj	15,728,640	7,259,457,280
model.language_model.layers.27.self_attn.q_norm	256	7,259,457,536
model.language_model.layers.27.self_attn.q_proj	15,728,640	7,275,186,176
model.language_model.layers.27.self_attn.v_proj	7,864,320	7,283,050,496
model.language_model.layers.28.input_layernorm	3,840	7,283,054,336
model.language_model.layers.28.mlp.down_proj	58,982,400	7,342,036,736
model.language_model.layers.28.mlp.gate_proj	58,982,400	7,401,019,136
model.language_model.layers.28.mlp.up_proj	58,982,400	7,460,001,536
model.language_model.layers.28.post_attention_layernorm	3,840	7,460,005,376
model.language_model.layers.28.post_feedforward_layernorm	3,840	7,460,009,216
model.language_model.layers.28.pre_feedforward_layernorm	3,840	7,460,013,056
model.language_model.layers.28.self_attn.k_norm	256	7,460,013,312
model.language_model.layers.28.self_attn.k_proj	7,864,320	7,467,877,632
model.language_model.layers.28.self_attn.o_proj	15,728,640	7,483,606,272
model.language_model.layers.28.self_attn.q_norm	256	7,483,606,528
model.language_model.layers.28.self_attn.q_proj	15,728,640	7,499,335,168
model.language_model.layers.28.self_attn.v_proj	7,864,320	7,507,199,488
model.language_model.layers.29.input_layernorm	3,840	7,507,203,328
model.language_model.layers.29.mlp.down_proj	58,982,400	7,566,185,728
model.language_model.layers.29.mlp.gate_proj	58,982,400	7,625,168,128
model.language_model.layers.29.mlp.up_proj	58,982,400	7,684,150,528
model.language_model.layers.29.post_attention_layernorm	3,840	7,684,154,368
model.language_model.layers.29.post_feedforward_layernorm	3,840	7,684,158,208
model.language_model.layers.29.pre_feedforward_layernorm	3,840	7,684,162,048
model.language_model.layers.29.self_attn.k_norm	256	7,684,162,304
model.language_model.layers.29.self_attn.k_proj	7,864,320	7,692,026,624
model.language_model.layers.29.self_attn.o_proj	15,728,640	7,707,755,264
model.language_model.layers.29.self_attn.q_norm	256	7,707,755,520
model.language_model.layers.29.self_attn.q_proj	15,728,640	7,723,484,160
model.language_model.layers.29.self_attn.v_proj	7,864,320	7,731,348,480
model.language_model.layers.30.input_layernorm	3,840	7,731,352,320
model.language_model.layers.30.mlp.down_proj	58,982,400	7,790,334,720
model.language_model.layers.30.mlp.gate_proj	58,982,400	7,849,317,120
model.language_model.layers.30.mlp.up_proj	58,982,400	7,908,299,520
model.language_model.layers.30.post_attention_layernorm	3,840	7,908,303,360

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Module	Params	Running total
model.language_model.layers.30. post_feedforward_layernorm	3,840	7,908,307,200
model.language_model.layers.30. pre_feedforward_layernorm	3,840	7,908,311,040
model.language_model.layers.30.self_attn.k_norm	256	7,908,311,296
model.language_model.layers.30.self_attn.k_proj	7,864,320	7,916,175,616
model.language_model.layers.30.self_attn.o_proj	15,728,640	7,931,904,256
model.language_model.layers.30.self_attn.q_norm	256	7,931,904,512
model.language_model.layers.30.self_attn.q_proj	15,728,640	7,947,633,152
model.language_model.layers.30.self_attn.v_proj	7,864,320	7,955,497,472
model.language_model.layers.31.input_layernorm	3,840	7,955,501,312
model.language_model.layers.31.mlp.down_proj	58,982,400	8,014,483,712
model.language_model.layers.31.mlp.gate_proj	58,982,400	8,073,466,112
model.language_model.layers.31.mlp.up_proj	58,982,400	8,132,448,512
model.language_model.layers.31. post_attention_layernorm	3,840	8,132,452,352
model.language_model.layers.31. post_feedforward_layernorm	3,840	8,132,456,192
model.language_model.layers.31. pre_feedforward_layernorm	3,840	8,132,460,032
model.language_model.layers.31.self_attn.k_norm	256	8,132,460,288
model.language_model.layers.31.self_attn.k_proj	7,864,320	8,140,324,608
model.language_model.layers.31.self_attn.o_proj	15,728,640	8,156,053,248
model.language_model.layers.31.self_attn.q_norm	256	8,156,053,504
model.language_model.layers.31.self_attn.q_proj	15,728,640	8,171,782,144
model.language_model.layers.31.self_attn.v_proj	7,864,320	8,179,646,464
model.language_model.layers.32.input_layernorm	3,840	8,179,650,304
model.language_model.layers.32.mlp.down_proj	58,982,400	8,238,632,704
model.language_model.layers.32.mlp.gate_proj	58,982,400	8,297,615,104
model.language_model.layers.32.mlp.up_proj	58,982,400	8,356,597,504
model.language_model.layers.32. post_attention_layernorm	3,840	8,356,601,344
model.language_model.layers.32. post_feedforward_layernorm	3,840	8,356,605,184
model.language_model.layers.32. pre_feedforward_layernorm	3,840	8,356,609,024
model.language_model.layers.32.self_attn.k_norm	256	8,356,609,280
model.language_model.layers.32.self_attn.k_proj	7,864,320	8,364,473,600
model.language_model.layers.32.self_attn.o_proj	15,728,640	8,380,202,240
model.language_model.layers.32.self_attn.q_norm	256	8,380,202,496
model.language_model.layers.32.self_attn.q_proj	15,728,640	8,395,931,136
model.language_model.layers.32.self_attn.v_proj	7,864,320	8,403,795,456
model.language_model.layers.33.input_layernorm	3,840	8,403,799,296
model.language_model.layers.33.mlp.down_proj	58,982,400	8,462,781,696
model.language_model.layers.33.mlp.gate_proj	58,982,400	8,521,764,096
model.language_model.layers.33.mlp.up_proj	58,982,400	8,580,746,496
model.language_model.layers.33. post_attention_layernorm	3,840	8,580,750,336
model.language_model.layers.33. post_feedforward_layernorm	3,840	8,580,754,176
model.language_model.layers.33. pre_feedforward_layernorm	3,840	8,580,758,016
model.language_model.layers.33.self_attn.k_norm	256	8,580,758,272
model.language_model.layers.33.self_attn.k_proj	7,864,320	8,588,622,592
model.language_model.layers.33.self_attn.o_proj	15,728,640	8,604,351,232
model.language_model.layers.33.self_attn.q_norm	256	8,604,351,488
model.language_model.layers.33.self_attn.q_proj	15,728,640	8,620,080,128
model.language_model.layers.33.self_attn.v_proj	7,864,320	8,627,944,448

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Module	Params	Running total
model.language_model.layers.34.input_layernorm	3,840	8,627,948,288
model.language_model.layers.34.mlp.down_proj	58,982,400	8,686,930,688
model.language_model.layers.34.mlp.gate_proj	58,982,400	8,745,913,088
model.language_model.layers.34.mlp.up_proj	58,982,400	8,804,895,488
model.language_model.layers.34.post_attention_layernorm	3,840	8,804,899,328
model.language_model.layers.34.post_feedforward_layernorm	3,840	8,804,903,168
model.language_model.layers.34.pre_feedforward_layernorm	3,840	8,804,907,008
model.language_model.layers.34.self_attn.k_norm	256	8,804,907,264
model.language_model.layers.34.self_attn.k_proj	7,864,320	8,812,771,584
model.language_model.layers.34.self_attn.o_proj	15,728,640	8,828,500,224
model.language_model.layers.34.self_attn.q_norm	256	8,828,500,480
model.language_model.layers.34.self_attn.q_proj	15,728,640	8,844,229,120
model.language_model.layers.34.self_attn.v_proj	7,864,320	8,852,093,440
model.language_model.layers.35.input_layernorm	3,840	8,852,097,280
model.language_model.layers.35.mlp.down_proj	58,982,400	8,911,079,680
model.language_model.layers.35.mlp.gate_proj	58,982,400	8,970,062,080
model.language_model.layers.35.mlp.up_proj	58,982,400	9,029,044,480
model.language_model.layers.35.post_attention_layernorm	3,840	9,029,048,320
model.language_model.layers.35.post_feedforward_layernorm	3,840	9,029,052,160
model.language_model.layers.35.pre_feedforward_layernorm	3,840	9,029,056,000
model.language_model.layers.35.self_attn.k_norm	256	9,029,056,256
model.language_model.layers.35.self_attn.k_proj	7,864,320	9,036,920,576
model.language_model.layers.35.self_attn.o_proj	15,728,640	9,052,649,216
model.language_model.layers.35.self_attn.q_norm	256	9,052,649,472
model.language_model.layers.35.self_attn.q_proj	15,728,640	9,068,378,112
model.language_model.layers.35.self_attn.v_proj	7,864,320	9,076,242,432
model.language_model.layers.36.input_layernorm	3,840	9,076,246,272
model.language_model.layers.36.mlp.down_proj	58,982,400	9,135,228,672
model.language_model.layers.36.mlp.gate_proj	58,982,400	9,194,211,072
model.language_model.layers.36.mlp.up_proj	58,982,400	9,253,193,472
model.language_model.layers.36.post_attention_layernorm	3,840	9,253,197,312
model.language_model.layers.36.post_feedforward_layernorm	3,840	9,253,201,152
model.language_model.layers.36.pre_feedforward_layernorm	3,840	9,253,204,992
model.language_model.layers.36.self_attn.k_norm	256	9,253,205,248
model.language_model.layers.36.self_attn.k_proj	7,864,320	9,261,069,568
model.language_model.layers.36.self_attn.o_proj	15,728,640	9,276,798,208
model.language_model.layers.36.self_attn.q_norm	256	9,276,798,464
model.language_model.layers.36.self_attn.q_proj	15,728,640	9,292,527,104
model.language_model.layers.36.self_attn.v_proj	7,864,320	9,300,391,424
model.language_model.layers.37.input_layernorm	3,840	9,300,395,264
model.language_model.layers.37.mlp.down_proj	58,982,400	9,359,377,664
model.language_model.layers.37.mlp.gate_proj	58,982,400	9,418,360,064
model.language_model.layers.37.mlp.up_proj	58,982,400	9,477,342,464
model.language_model.layers.37.post_attention_layernorm	3,840	9,477,346,304
model.language_model.layers.37.post_feedforward_layernorm	3,840	9,477,350,144
model.language_model.layers.37.pre_feedforward_layernorm	3,840	9,477,353,984

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Module	Params	Running total
model.language_model.layers.37.self_attn.k_norm	256	9,477,354,240
model.language_model.layers.37.self_attn.k_proj	7,864,320	9,485,218,560
model.language_model.layers.37.self_attn.o_proj	15,728,640	9,500,947,200
model.language_model.layers.37.self_attn.q_norm	256	9,500,947,456
model.language_model.layers.37.self_attn.q_proj	15,728,640	9,516,676,096
model.language_model.layers.37.self_attn.v_proj	7,864,320	9,524,540,416
model.language_model.layers.38.input_layernorm	3,840	9,524,544,256
model.language_model.layers.38.mlp.down_proj	58,982,400	9,583,526,656
model.language_model.layers.38.mlp.gate_proj	58,982,400	9,642,509,056
model.language_model.layers.38.mlp.up_proj	58,982,400	9,701,491,456
model.language_model.layers.38.post_attention_layernorm	3,840	9,701,495,296
model.language_model.layers.38.post_feedforward_layernorm	3,840	9,701,499,136
model.language_model.layers.38.pre_feedforward_layernorm	3,840	9,701,502,976
model.language_model.layers.38.self_attn.k_norm	256	9,701,503,232
model.language_model.layers.38.self_attn.k_proj	7,864,320	9,709,367,552
model.language_model.layers.38.self_attn.o_proj	15,728,640	9,725,096,192
model.language_model.layers.38.self_attn.q_norm	256	9,725,096,448
model.language_model.layers.38.self_attn.q_proj	15,728,640	9,740,825,088
model.language_model.layers.38.self_attn.v_proj	7,864,320	9,748,689,408
model.language_model.layers.39.input_layernorm	3,840	9,748,693,248
model.language_model.layers.39.mlp.down_proj	58,982,400	9,807,675,648
model.language_model.layers.39.mlp.gate_proj	58,982,400	9,866,658,048
model.language_model.layers.39.mlp.up_proj	58,982,400	9,925,640,448
model.language_model.layers.39.post_attention_layernorm	3,840	9,925,644,288
model.language_model.layers.39.post_feedforward_layernorm	3,840	9,925,648,128
model.language_model.layers.39.pre_feedforward_layernorm	3,840	9,925,651,968
model.language_model.layers.39.self_attn.k_norm	256	9,925,652,224
model.language_model.layers.39.self_attn.k_proj	7,864,320	9,933,516,544
model.language_model.layers.39.self_attn.o_proj	15,728,640	9,949,245,184
model.language_model.layers.39.self_attn.q_norm	256	9,949,245,440
model.language_model.layers.39.self_attn.q_proj	15,728,640	9,964,974,080
model.language_model.layers.39.self_attn.v_proj	7,864,320	9,972,838,400
model.language_model.layers.40.input_layernorm	3,840	9,972,842,240
model.language_model.layers.40.mlp.down_proj	58,982,400	10,031,824,640
model.language_model.layers.40.mlp.gate_proj	58,982,400	10,090,807,040
model.language_model.layers.40.mlp.up_proj	58,982,400	10,149,789,440
model.language_model.layers.40.post_attention_layernorm	3,840	10,149,793,280
model.language_model.layers.40.post_feedforward_layernorm	3,840	10,149,797,120
model.language_model.layers.40.pre_feedforward_layernorm	3,840	10,149,800,960
model.language_model.layers.40.self_attn.k_norm	256	10,149,801,216
model.language_model.layers.40.self_attn.k_proj	7,864,320	10,157,665,536
model.language_model.layers.40.self_attn.o_proj	15,728,640	10,173,394,176
model.language_model.layers.40.self_attn.q_norm	256	10,173,394,432
model.language_model.layers.40.self_attn.q_proj	15,728,640	10,189,123,072
model.language_model.layers.40.self_attn.v_proj	7,864,320	10,196,987,392
model.language_model.layers.41.input_layernorm	3,840	10,196,991,232
model.language_model.layers.41.mlp.down_proj	58,982,400	10,255,973,632
model.language_model.layers.41.mlp.gate_proj	58,982,400	10,314,956,032
model.language_model.layers.41.mlp.up_proj	58,982,400	10,373,938,432

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Module	Params	Running total
model.language_model.layers.41.post_attention_layernorm	3,840	10,373,942,272
model.language_model.layers.41.post_feedforward_layernorm	3,840	10,373,946,112
model.language_model.layers.41.pre_feedforward_layernorm	3,840	10,373,949,952
model.language_model.layers.41.self_attn.k_norm	256	10,373,950,208
model.language_model.layers.41.self_attn.k_proj	7,864,320	10,381,814,528
model.language_model.layers.41.self_attn.o_proj	15,728,640	10,397,543,168
model.language_model.layers.41.self_attn.q_norm	256	10,397,543,424
model.language_model.layers.41.self_attn.q_proj	15,728,640	10,413,272,064
model.language_model.layers.41.self_attn.v_proj	7,864,320	10,421,136,384
model.language_model.layers.42.input_layernorm	3,840	10,421,140,224
model.language_model.layers.42.mlp.down_proj	58,982,400	10,480,122,624
model.language_model.layers.42.mlp.gate_proj	58,982,400	10,539,105,024
model.language_model.layers.42.mlp.up_proj	58,982,400	10,598,087,424
model.language_model.layers.42.post_attention_layernorm	3,840	10,598,091,264
model.language_model.layers.42.post_feedforward_layernorm	3,840	10,598,095,104
model.language_model.layers.42.pre_feedforward_layernorm	3,840	10,598,098,944
model.language_model.layers.42.self_attn.k_norm	256	10,598,099,200
model.language_model.layers.42.self_attn.k_proj	7,864,320	10,605,963,520
model.language_model.layers.42.self_attn.o_proj	15,728,640	10,621,692,160
model.language_model.layers.42.self_attn.q_norm	256	10,621,692,416
model.language_model.layers.42.self_attn.q_proj	15,728,640	10,637,421,056
model.language_model.layers.42.self_attn.v_proj	7,864,320	10,645,285,376
model.language_model.layers.43.input_layernorm	3,840	10,645,289,216
model.language_model.layers.43.mlp.down_proj	58,982,400	10,704,271,616
model.language_model.layers.43.mlp.gate_proj	58,982,400	10,763,254,016
model.language_model.layers.43.mlp.up_proj	58,982,400	10,822,236,416
model.language_model.layers.43.post_attention_layernorm	3,840	10,822,240,256
model.language_model.layers.43.post_feedforward_layernorm	3,840	10,822,244,096
model.language_model.layers.43.pre_feedforward_layernorm	3,840	10,822,247,936
model.language_model.layers.43.self_attn.k_norm	256	10,822,248,192
model.language_model.layers.43.self_attn.k_proj	7,864,320	10,830,112,512
model.language_model.layers.43.self_attn.o_proj	15,728,640	10,845,841,152
model.language_model.layers.43.self_attn.q_norm	256	10,845,841,408
model.language_model.layers.43.self_attn.q_proj	15,728,640	10,861,570,048
model.language_model.layers.43.self_attn.v_proj	7,864,320	10,869,434,368
model.language_model.layers.44.input_layernorm	3,840	10,869,438,208
model.language_model.layers.44.mlp.down_proj	58,982,400	10,928,420,608
model.language_model.layers.44.mlp.gate_proj	58,982,400	10,987,403,008
model.language_model.layers.44.mlp.up_proj	58,982,400	11,046,385,408
model.language_model.layers.44.post_attention_layernorm	3,840	11,046,389,248
model.language_model.layers.44.post_feedforward_layernorm	3,840	11,046,393,088
model.language_model.layers.44.pre_feedforward_layernorm	3,840	11,046,396,928
model.language_model.layers.44.self_attn.k_norm	256	11,046,397,184
model.language_model.layers.44.self_attn.k_proj	7,864,320	11,054,261,504
model.language_model.layers.44.self_attn.o_proj	15,728,640	11,069,990,144
model.language_model.layers.44.self_attn.q_norm	256	11,069,990,400

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Module	Params	Running total
model.language_model.layers.44.self_attn.q_proj	15,728,640	11,085,719,040
model.language_model.layers.44.self_attn.v_proj	7,864,320	11,093,583,360
model.language_model.layers.45.input_layernorm	3,840	11,093,587,200
model.language_model.layers.45.mlp.down_proj	58,982,400	11,152,569,600
model.language_model.layers.45.mlp.gate_proj	58,982,400	11,211,552,000
model.language_model.layers.45.mlp.up_proj	58,982,400	11,270,534,400
model.language_model.layers.45.post_attention_layernorm	3,840	11,270,538,240
model.language_model.layers.45.post_feedforward_layernorm	3,840	11,270,542,080
model.language_model.layers.45.pre_feedforward_layernorm	3,840	11,270,545,920
model.language_model.layers.45.self_attn.k_norm	256	11,270,546,176
model.language_model.layers.45.self_attn.k_proj	7,864,320	11,278,410,496
model.language_model.layers.45.self_attn.o_proj	15,728,640	11,294,139,136
model.language_model.layers.45.self_attn.q_norm	256	11,294,139,392
model.language_model.layers.45.self_attn.q_proj	15,728,640	11,309,868,032
model.language_model.layers.45.self_attn.v_proj	7,864,320	11,317,732,352
model.language_model.layers.46.input_layernorm	3,840	11,317,736,192
model.language_model.layers.46.mlp.down_proj	58,982,400	11,376,718,592
model.language_model.layers.46.mlp.gate_proj	58,982,400	11,435,700,992
model.language_model.layers.46.mlp.up_proj	58,982,400	11,494,683,392
model.language_model.layers.46.post_attention_layernorm	3,840	11,494,687,232
model.language_model.layers.46.post_feedforward_layernorm	3,840	11,494,691,072
model.language_model.layers.46.pre_feedforward_layernorm	3,840	11,494,694,912
model.language_model.layers.46.self_attn.k_norm	256	11,494,695,168
model.language_model.layers.46.self_attn.k_proj	7,864,320	11,502,559,488
model.language_model.layers.46.self_attn.o_proj	15,728,640	11,518,288,128
model.language_model.layers.46.self_attn.q_norm	256	11,518,288,384
model.language_model.layers.46.self_attn.q_proj	15,728,640	11,534,017,024
model.language_model.layers.46.self_attn.v_proj	7,864,320	11,541,881,344
model.language_model.layers.47.input_layernorm	3,840	11,541,885,184
model.language_model.layers.47.mlp.down_proj	58,982,400	11,600,867,584
model.language_model.layers.47.mlp.gate_proj	58,982,400	11,659,849,984
model.language_model.layers.47.mlp.up_proj	58,982,400	11,718,832,384
model.language_model.layers.47.post_attention_layernorm	3,840	11,718,836,224
model.language_model.layers.47.post_feedforward_layernorm	3,840	11,718,840,064
model.language_model.layers.47.pre_feedforward_layernorm	3,840	11,718,843,904
model.language_model.layers.47.self_attn.k_norm	256	11,718,844,160
model.language_model.layers.47.self_attn.k_proj	7,864,320	11,726,708,480
model.language_model.layers.47.self_attn.o_proj	15,728,640	11,742,437,120
model.language_model.layers.47.self_attn.q_norm	256	11,742,437,376
model.language_model.layers.47.self_attn.q_proj	15,728,640	11,758,166,016
model.language_model.layers.47.self_attn.v_proj	7,864,320	11,766,030,336
model.language_model.norm	3,840	11,766,034,176
model.multi_modal_projector	4,423,680	11,770,457,856
model.multi_modal_projector.mm_soft_emb_norm	1,152	11,770,459,008
model.vision_tower.embeddings.patch_embedding	678,528	11,771,137,536
model.vision_tower.embeddings.position_embedding	4,718,592	11,775,856,128
model.vision_tower.encoder.layers.0.layer_norm1	2,304	11,775,858,432
model.vision_tower.encoder.layers.0.layer_norm2	2,304	11,775,860,736
model.vision_tower.encoder.layers.0.mlp.fc1	4,962,512	11,780,823,248

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Module	Params	Running total
model.vision_tower.encoder.layers.0.mlp.fc2	4,959,360	11,785,782,608
model.vision_tower.encoder.layers.0.self_attn.k_proj	1,328,256	11,787,110,864
model.vision_tower.encoder.layers.0.self_attn.out_proj	1,328,256	11,788,439,120
model.vision_tower.encoder.layers.0.self_attn.q_proj	1,328,256	11,789,767,376
model.vision_tower.encoder.layers.0.self_attn.v_proj	1,328,256	11,791,095,632
model.vision_tower.encoder.layers.1.layer_norm1	2,304	11,791,097,936
model.vision_tower.encoder.layers.1.layer_norm2	2,304	11,791,100,240
model.vision_tower.encoder.layers.1.mlp.fc1	4,962,512	11,796,062,752
model.vision_tower.encoder.layers.1.mlp.fc2	4,959,360	11,801,022,112
model.vision_tower.encoder.layers.1.self_attn.k_proj	1,328,256	11,802,350,368
model.vision_tower.encoder.layers.1.self_attn.out_proj	1,328,256	11,803,678,624
model.vision_tower.encoder.layers.1.self_attn.q_proj	1,328,256	11,805,006,880
model.vision_tower.encoder.layers.1.self_attn.v_proj	1,328,256	11,806,335,136
model.vision_tower.encoder.layers.2.layer_norm1	2,304	11,806,337,440
model.vision_tower.encoder.layers.2.layer_norm2	2,304	11,806,339,744
model.vision_tower.encoder.layers.2.mlp.fc1	4,962,512	11,811,302,256
model.vision_tower.encoder.layers.2.mlp.fc2	4,959,360	11,816,261,616
model.vision_tower.encoder.layers.2.self_attn.k_proj	1,328,256	11,817,589,872
model.vision_tower.encoder.layers.2.self_attn.out_proj	1,328,256	11,818,918,128
model.vision_tower.encoder.layers.2.self_attn.q_proj	1,328,256	11,820,246,384
model.vision_tower.encoder.layers.2.self_attn.v_proj	1,328,256	11,821,574,640
model.vision_tower.encoder.layers.3.layer_norm1	2,304	11,821,576,944
model.vision_tower.encoder.layers.3.layer_norm2	2,304	11,821,579,248
model.vision_tower.encoder.layers.3.mlp.fc1	4,962,512	11,826,541,760
model.vision_tower.encoder.layers.3.mlp.fc2	4,959,360	11,831,501,120
model.vision_tower.encoder.layers.3.self_attn.k_proj	1,328,256	11,832,829,376
model.vision_tower.encoder.layers.3.self_attn.out_proj	1,328,256	11,834,157,632
model.vision_tower.encoder.layers.3.self_attn.q_proj	1,328,256	11,835,485,888
model.vision_tower.encoder.layers.3.self_attn.v_proj	1,328,256	11,836,814,144
model.vision_tower.encoder.layers.4.layer_norm1	2,304	11,836,816,448
model.vision_tower.encoder.layers.4.layer_norm2	2,304	11,836,818,752
model.vision_tower.encoder.layers.4.mlp.fc1	4,962,512	11,841,781,264
model.vision_tower.encoder.layers.4.mlp.fc2	4,959,360	11,846,740,624
model.vision_tower.encoder.layers.4.self_attn.k_proj	1,328,256	11,848,068,880
model.vision_tower.encoder.layers.4.self_attn.out_proj	1,328,256	11,849,397,136
model.vision_tower.encoder.layers.4.self_attn.q_proj	1,328,256	11,850,725,392
model.vision_tower.encoder.layers.4.self_attn.v_proj	1,328,256	11,852,053,648
model.vision_tower.encoder.layers.5.layer_norm1	2,304	11,852,055,952

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Module	Params	Running total
model.vision_tower.encoder.layers.5.layer_norm2	2,304	11,852,058,256
model.vision_tower.encoder.layers.5.mlp.fc1	4,962,512	11,857,020,768
model.vision_tower.encoder.layers.5.mlp.fc2	4,959,360	11,861,980,128
model.vision_tower.encoder.layers.5.self_attn.k_proj	1,328,256	11,863,308,384
model.vision_tower.encoder.layers.5.self_attn.out_proj	1,328,256	11,864,636,640
model.vision_tower.encoder.layers.5.self_attn.q_proj	1,328,256	11,865,964,896
model.vision_tower.encoder.layers.5.self_attn.v_proj	1,328,256	11,867,293,152
model.vision_tower.encoder.layers.6.layer_norm1	2,304	11,867,295,456
model.vision_tower.encoder.layers.6.layer_norm2	2,304	11,867,297,760
model.vision_tower.encoder.layers.6.mlp.fc1	4,962,512	11,872,260,272
model.vision_tower.encoder.layers.6.mlp.fc2	4,959,360	11,877,219,632
model.vision_tower.encoder.layers.6.self_attn.k_proj	1,328,256	11,878,547,888
model.vision_tower.encoder.layers.6.self_attn.out_proj	1,328,256	11,879,876,144
model.vision_tower.encoder.layers.6.self_attn.q_proj	1,328,256	11,881,204,400
model.vision_tower.encoder.layers.6.self_attn.v_proj	1,328,256	11,882,532,656
model.vision_tower.encoder.layers.7.layer_norm1	2,304	11,882,534,960
model.vision_tower.encoder.layers.7.layer_norm2	2,304	11,882,537,264
model.vision_tower.encoder.layers.7.mlp.fc1	4,962,512	11,887,499,776
model.vision_tower.encoder.layers.7.mlp.fc2	4,959,360	11,892,459,136
model.vision_tower.encoder.layers.7.self_attn.k_proj	1,328,256	11,893,787,392
model.vision_tower.encoder.layers.7.self_attn.out_proj	1,328,256	11,895,115,648
model.vision_tower.encoder.layers.7.self_attn.q_proj	1,328,256	11,896,443,904
model.vision_tower.encoder.layers.7.self_attn.v_proj	1,328,256	11,897,772,160
model.vision_tower.encoder.layers.8.layer_norm1	2,304	11,897,774,464
model.vision_tower.encoder.layers.8.layer_norm2	2,304	11,897,776,768
model.vision_tower.encoder.layers.8.mlp.fc1	4,962,512	11,902,739,280
model.vision_tower.encoder.layers.8.mlp.fc2	4,959,360	11,907,698,640
model.vision_tower.encoder.layers.8.self_attn.k_proj	1,328,256	11,909,026,896
model.vision_tower.encoder.layers.8.self_attn.out_proj	1,328,256	11,910,355,152
model.vision_tower.encoder.layers.8.self_attn.q_proj	1,328,256	11,911,683,408
model.vision_tower.encoder.layers.8.self_attn.v_proj	1,328,256	11,913,011,664
model.vision_tower.encoder.layers.9.layer_norm1	2,304	11,913,013,968
model.vision_tower.encoder.layers.9.layer_norm2	2,304	11,913,016,272
model.vision_tower.encoder.layers.9.mlp.fc1	4,962,512	11,917,978,784
model.vision_tower.encoder.layers.9.mlp.fc2	4,959,360	11,922,938,144
model.vision_tower.encoder.layers.9.self_attn.k_proj	1,328,256	11,924,266,400
model.vision_tower.encoder.layers.9.self_attn.out_proj	1,328,256	11,925,594,656
model.vision_tower.encoder.layers.9.self_attn.q_proj	1,328,256	11,926,922,912

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Module	Params	Running total
model.vision_tower.encoder.layers.9.self_attn.v_proj	1,328,256	11,928,251,168
model.vision_tower.encoder.layers.10.layer_norm1	2,304	11,928,253,472
model.vision_tower.encoder.layers.10.layer_norm2	2,304	11,928,255,776
model.vision_tower.encoder.layers.10.mlp.fc1	4,962,512	11,933,218,288
model.vision_tower.encoder.layers.10.mlp.fc2	4,959,360	11,938,177,648
model.vision_tower.encoder.layers.10.self_attn.k_proj	1,328,256	11,939,505,904
model.vision_tower.encoder.layers.10.self_attn.out_proj	1,328,256	11,940,834,160
model.vision_tower.encoder.layers.10.self_attn.q_proj	1,328,256	11,942,162,416
model.vision_tower.encoder.layers.10.self_attn.v_proj	1,328,256	11,943,490,672
model.vision_tower.encoder.layers.11.layer_norm1	2,304	11,943,492,976
model.vision_tower.encoder.layers.11.layer_norm2	2,304	11,943,495,280
model.vision_tower.encoder.layers.11.mlp.fc1	4,962,512	11,948,457,792
model.vision_tower.encoder.layers.11.mlp.fc2	4,959,360	11,953,417,152
model.vision_tower.encoder.layers.11.self_attn.k_proj	1,328,256	11,954,745,408
model.vision_tower.encoder.layers.11.self_attn.out_proj	1,328,256	11,956,073,664
model.vision_tower.encoder.layers.11.self_attn.q_proj	1,328,256	11,957,401,920
model.vision_tower.encoder.layers.11.self_attn.v_proj	1,328,256	11,958,730,176
model.vision_tower.encoder.layers.12.layer_norm1	2,304	11,958,732,480
model.vision_tower.encoder.layers.12.layer_norm2	2,304	11,958,734,784
model.vision_tower.encoder.layers.12.mlp.fc1	4,962,512	11,963,697,296
model.vision_tower.encoder.layers.12.mlp.fc2	4,959,360	11,968,656,656
model.vision_tower.encoder.layers.12.self_attn.k_proj	1,328,256	11,969,984,912
model.vision_tower.encoder.layers.12.self_attn.out_proj	1,328,256	11,971,313,168
model.vision_tower.encoder.layers.12.self_attn.q_proj	1,328,256	11,972,641,424
model.vision_tower.encoder.layers.12.self_attn.v_proj	1,328,256	11,973,969,680
model.vision_tower.encoder.layers.13.layer_norm1	2,304	11,973,971,984
model.vision_tower.encoder.layers.13.layer_norm2	2,304	11,973,974,288
model.vision_tower.encoder.layers.13.mlp.fc1	4,962,512	11,978,936,800
model.vision_tower.encoder.layers.13.mlp.fc2	4,959,360	11,983,896,160
model.vision_tower.encoder.layers.13.self_attn.k_proj	1,328,256	11,985,224,416
model.vision_tower.encoder.layers.13.self_attn.out_proj	1,328,256	11,986,552,672
model.vision_tower.encoder.layers.13.self_attn.q_proj	1,328,256	11,987,880,928
model.vision_tower.encoder.layers.13.self_attn.v_proj	1,328,256	11,989,209,184
model.vision_tower.encoder.layers.14.layer_norm1	2,304	11,989,211,488
model.vision_tower.encoder.layers.14.layer_norm2	2,304	11,989,213,792
model.vision_tower.encoder.layers.14.mlp.fc1	4,962,512	11,994,176,304
model.vision_tower.encoder.layers.14.mlp.fc2	4,959,360	11,999,135,664
model.vision_tower.encoder.layers.14.self_attn.k_proj	1,328,256	12,000,463,920
model.vision_tower.encoder.layers.14.self_attn.out_proj	1,328,256	12,001,792,176

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Module	Params	Running total
model.vision_tower.encoder.layers.14.self_attn.q_proj	1,328,256	12,003,120,432
model.vision_tower.encoder.layers.14.self_attn.v_proj	1,328,256	12,004,448,688
model.vision_tower.encoder.layers.15.layer_norm1	2,304	12,004,450,992
model.vision_tower.encoder.layers.15.layer_norm2	2,304	12,004,453,296
model.vision_tower.encoder.layers.15.mlp.fc1	4,962,512	12,009,415,808
model.vision_tower.encoder.layers.15.mlp.fc2	4,959,360	12,014,375,168
model.vision_tower.encoder.layers.15.self_attn.k_proj	1,328,256	12,015,703,424
model.vision_tower.encoder.layers.15.self_attn.out_proj	1,328,256	12,017,031,680
model.vision_tower.encoder.layers.15.self_attn.q_proj	1,328,256	12,018,359,936
model.vision_tower.encoder.layers.15.self_attn.v_proj	1,328,256	12,019,688,192
model.vision_tower.encoder.layers.16.layer_norm1	2,304	12,019,690,496
model.vision_tower.encoder.layers.16.layer_norm2	2,304	12,019,692,800
model.vision_tower.encoder.layers.16.mlp.fc1	4,962,512	12,024,655,312
model.vision_tower.encoder.layers.16.mlp.fc2	4,959,360	12,029,614,672
model.vision_tower.encoder.layers.16.self_attn.k_proj	1,328,256	12,030,942,928
model.vision_tower.encoder.layers.16.self_attn.out_proj	1,328,256	12,032,271,184
model.vision_tower.encoder.layers.16.self_attn.q_proj	1,328,256	12,033,599,440
model.vision_tower.encoder.layers.16.self_attn.v_proj	1,328,256	12,034,927,696
model.vision_tower.encoder.layers.17.layer_norm1	2,304	12,034,930,000
model.vision_tower.encoder.layers.17.layer_norm2	2,304	12,034,932,304
model.vision_tower.encoder.layers.17.mlp.fc1	4,962,512	12,039,894,816
model.vision_tower.encoder.layers.17.mlp.fc2	4,959,360	12,044,854,176
model.vision_tower.encoder.layers.17.self_attn.k_proj	1,328,256	12,046,182,432
model.vision_tower.encoder.layers.17.self_attn.out_proj	1,328,256	12,047,510,688
model.vision_tower.encoder.layers.17.self_attn.q_proj	1,328,256	12,048,838,944
model.vision_tower.encoder.layers.17.self_attn.v_proj	1,328,256	12,050,167,200
model.vision_tower.encoder.layers.18.layer_norm1	2,304	12,050,169,504
model.vision_tower.encoder.layers.18.layer_norm2	2,304	12,050,171,808
model.vision_tower.encoder.layers.18.mlp.fc1	4,962,512	12,055,134,320
model.vision_tower.encoder.layers.18.mlp.fc2	4,959,360	12,060,093,680
model.vision_tower.encoder.layers.18.self_attn.k_proj	1,328,256	12,061,421,936
model.vision_tower.encoder.layers.18.self_attn.out_proj	1,328,256	12,062,750,192
model.vision_tower.encoder.layers.18.self_attn.q_proj	1,328,256	12,064,078,448
model.vision_tower.encoder.layers.18.self_attn.v_proj	1,328,256	12,065,406,704
model.vision_tower.encoder.layers.19.layer_norm1	2,304	12,065,409,008
model.vision_tower.encoder.layers.19.layer_norm2	2,304	12,065,411,312
model.vision_tower.encoder.layers.19.mlp.fc1	4,962,512	12,070,373,824
model.vision_tower.encoder.layers.19.mlp.fc2	4,959,360	12,075,333,184
model.vision_tower.encoder.layers.19.self_attn.k_proj	1,328,256	12,076,661,440

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Module	Params	Running total
model.vision_tower.encoder.layers.19.self_attn.out_proj	1,328,256	12,077,989,696
model.vision_tower.encoder.layers.19.self_attn.q_proj	1,328,256	12,079,317,952
model.vision_tower.encoder.layers.19.self_attn.v_proj	1,328,256	12,080,646,208
model.vision_tower.encoder.layers.20.layer_norm1	2,304	12,080,648,512
model.vision_tower.encoder.layers.20.layer_norm2	2,304	12,080,650,816
model.vision_tower.encoder.layers.20.mlp.fc1	4,962,512	12,085,613,328
model.vision_tower.encoder.layers.20.mlp.fc2	4,959,360	12,090,572,688
model.vision_tower.encoder.layers.20.self_attn.k_proj	1,328,256	12,091,900,944
model.vision_tower.encoder.layers.20.self_attn.out_proj	1,328,256	12,093,229,200
model.vision_tower.encoder.layers.20.self_attn.q_proj	1,328,256	12,094,557,456
model.vision_tower.encoder.layers.20.self_attn.v_proj	1,328,256	12,095,885,712
model.vision_tower.encoder.layers.21.layer_norm1	2,304	12,095,888,016
model.vision_tower.encoder.layers.21.layer_norm2	2,304	12,095,890,320
model.vision_tower.encoder.layers.21.mlp.fc1	4,962,512	12,100,852,832
model.vision_tower.encoder.layers.21.mlp.fc2	4,959,360	12,105,812,192
model.vision_tower.encoder.layers.21.self_attn.k_proj	1,328,256	12,107,140,448
model.vision_tower.encoder.layers.21.self_attn.out_proj	1,328,256	12,108,468,704
model.vision_tower.encoder.layers.21.self_attn.q_proj	1,328,256	12,109,796,960
model.vision_tower.encoder.layers.21.self_attn.v_proj	1,328,256	12,111,125,216
model.vision_tower.encoder.layers.22.layer_norm1	2,304	12,111,127,520
model.vision_tower.encoder.layers.22.layer_norm2	2,304	12,111,129,824
model.vision_tower.encoder.layers.22.mlp.fc1	4,962,512	12,116,092,336
model.vision_tower.encoder.layers.22.mlp.fc2	4,959,360	12,121,051,696
model.vision_tower.encoder.layers.22.self_attn.k_proj	1,328,256	12,122,379,952
model.vision_tower.encoder.layers.22.self_attn.out_proj	1,328,256	12,123,708,208
model.vision_tower.encoder.layers.22.self_attn.q_proj	1,328,256	12,125,036,464
model.vision_tower.encoder.layers.22.self_attn.v_proj	1,328,256	12,126,364,720
model.vision_tower.encoder.layers.23.layer_norm1	2,304	12,126,367,024
model.vision_tower.encoder.layers.23.layer_norm2	2,304	12,126,369,328
model.vision_tower.encoder.layers.23.mlp.fc1	4,962,512	12,131,331,840
model.vision_tower.encoder.layers.23.mlp.fc2	4,959,360	12,136,291,200
model.vision_tower.encoder.layers.23.self_attn.k_proj	1,328,256	12,137,619,456
model.vision_tower.encoder.layers.23.self_attn.out_proj	1,328,256	12,138,947,712
model.vision_tower.encoder.layers.23.self_attn.q_proj	1,328,256	12,140,275,968
model.vision_tower.encoder.layers.23.self_attn.v_proj	1,328,256	12,141,604,224
model.vision_tower.encoder.layers.24.layer_norm1	2,304	12,141,606,528
model.vision_tower.encoder.layers.24.layer_norm2	2,304	12,141,608,832
model.vision_tower.encoder.layers.24.mlp.fc1	4,962,512	12,146,571,344
model.vision_tower.encoder.layers.24.mlp.fc2	4,959,360	12,151,530,704

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Module	Params	Running total
model.vision_tower.encoder.layers.24.self_attn.k_proj	1,328,256	12,152,858,960
model.vision_tower.encoder.layers.24.self_attn.out_proj	1,328,256	12,154,187,216
model.vision_tower.encoder.layers.24.self_attn.q_proj	1,328,256	12,155,515,472
model.vision_tower.encoder.layers.24.self_attn.v_proj	1,328,256	12,156,843,728
model.vision_tower.encoder.layers.25.layer_norm1	2,304	12,156,846,032
model.vision_tower.encoder.layers.25.layer_norm2	2,304	12,156,848,336
model.vision_tower.encoder.layers.25.mlp.fc1	4,962,512	12,161,810,848
model.vision_tower.encoder.layers.25.mlp.fc2	4,959,360	12,166,770,208
model.vision_tower.encoder.layers.25.self_attn.k_proj	1,328,256	12,168,098,464
model.vision_tower.encoder.layers.25.self_attn.out_proj	1,328,256	12,169,426,720
model.vision_tower.encoder.layers.25.self_attn.q_proj	1,328,256	12,170,754,976
model.vision_tower.encoder.layers.25.self_attn.v_proj	1,328,256	12,172,083,232
model.vision_tower.encoder.layers.26.layer_norm1	2,304	12,172,085,536
model.vision_tower.encoder.layers.26.layer_norm2	2,304	12,172,087,840
model.vision_tower.encoder.layers.26.mlp.fc1	4,962,512	12,177,050,352
model.vision_tower.encoder.layers.26.mlp.fc2	4,959,360	12,182,009,712
model.vision_tower.encoder.layers.26.self_attn.k_proj	1,328,256	12,183,337,968
model.vision_tower.encoder.layers.26.self_attn.out_proj	1,328,256	12,184,666,224
model.vision_tower.encoder.layers.26.self_attn.q_proj	1,328,256	12,185,994,480
model.vision_tower.encoder.layers.26.self_attn.v_proj	1,328,256	12,187,322,736
model.vision_tower.post_layernorm	2,304	12,187,325,040